

First-child housing loading without Stone-Geary

Bounded fixed-preference diagnostic; no recalibration or production adoption

Case	Parent housing weight	Fertility	Birth rooms	Childless (%)	Loss
Floor, linear benefit	0.267	2.100	0.785	18.279	285.640
Floor, concave benefit	0.267	2.042	0.783	19.206	296.264
No floor, no loading	0.267	2.633	0.120	3.267	3.71e+03
No floor, +10 points	0.367	2.605	0.941	3.737	2.87e+03
No floor, +20 points	0.467	2.540	1.538	4.972	2.08e+03

One-child benefit held fixed. New cases fix child-benefit curvature at 0.140; original control is linear.

Positive loading raises housing weight only when children are at home. Later children add no further share loading.

Earnings, initial wealth/income, mortality, bequests, credit, shocks, supply and fiscal rule are unchanged.

Changing housing weights uses the inherited fixed-reference-rent utility normalization; no estimated scale added.

Entry composition is held fixed. Reproduction gaps are reported rather than closed by refitting child benefits.

Birth rooms is the inherited stationary proxy, not a matched event study. Frozen old ACS targets retained for comparison.

The original floor response also misses the 1.465-room target. These fixed points cannot rank optimized specifications.

Specification and interpretation

All symbols below describe economic objects; numerical choices are separate

$$u(C, m) = \frac{C^{1-\sigma}}{1-\sigma} + \psi \frac{m^{1-\kappa}}{1-\kappa}$$

$$C = A(m) \frac{c^{\alpha(m)} s^{1-\alpha(m)}}{e(m)}$$

$$\alpha(m) = \alpha_0 - \lambda \mathbf{1}\{m > 0\}, \quad e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}$$

c: nonhousing consumption; s: housing services, including the unchanged owner premium; m: children at home.

sigma: material-consumption curvature (2 here); kappa: child-benefit curvature (0.140 in new cases).

psi weights enjoyment of children relative to material consumption; C is the household material-consumption index.

alpha_0 = 0.733 is the childless consumption weight; lambda is the positive first-child housing loading.

e(m) is the household needs scale: 1.000 without children, about 1.234 with one and 1.450 with two.

A(m) is a substantive fixed normalization, not an estimated parameter. For uncapped renters at the reference rent,

equal expenditure attains the same material index before household needs. This does not extend to all owners.

The benefit code uses $b \cdot m^{(1-\kappa)}$. Displayed $\psi = (1-\kappa) \cdot b$, so changing curvature preserves b at m=1.

Housing shares increase from 0.267 without children to 0.367 or 0.467 with children in the positive-loading cases.

Fertility incentives and demographic closure

Positive gap means trying now is preferred to waiting before the current taste draw

Case	First births at positive gaps (%)	Later births at positive gaps (%)	Zero first-birth attempts (% risk)	Replacement gap (%)
Floor, linear benefit	54.654	29.763	4.872	7.97e-06
Floor, concave benefit	53.344	25.880	4.787	2.767
No floor, no loading	73.328	23.615	5.24e-08	20.251
No floor, +10 points	67.316	22.934	5.69e-08	19.401
No floor, +20 points	59.055	22.064	5.27e-08	17.327

Continuation values retain future taste uncertainty. This is not a zero-shock equilibrium or a lifetime childlessness comparison.

Zero probabilities combine infeasibility and underflow. Changes in weighted shares also reflect population composition.

Replacement gaps are reported at fixed child benefit and fixed normalized entry; no closed-renewal condition is imposed.

Floor, linear benefit: complete target fit

Same target definitions and weights across cases; normalization row is descriptive at fixed child benefit

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.100	2.100	1.67e-07		
cps_childlessness	0.198	0.183	-0.015	3.55e+04	8.528
cps_exactly_one	0.214	0.231	0.018	2.7e+04	8.350
nchs_mean_age	25.976	26.324	0.348	139.828	16.896
nchs_share30	0.249	0.246	-0.003	1.39e+04	0.149
wealth_earnings	6.927	6.215	-0.711	7.595	3.844
bequest_wealth	0.007	0.007	-0.000407	5.17e+06	0.854
old_dispersion	3.516	2.910	-0.606	10.616	3.892
mean_rooms	5.608	6.543	0.935	128.021	111.867
ownership_30_55	0.676	0.762	0.086	2.34e+03	17.120
first_birth_rooms	1.465	0.785	-0.680	137.565	63.657
family_rooms	0.385	0.220	-0.165	280.528	7.636
recent_parent_ownership	0.128	0.088	-0.040	2.71e+04	42.846

Floor, linear benefit: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
H0	8.646	0.200	80.000	False	held at reference estimate; no recalibration
beta annual	0.962	0.940	0.990	False	held at reference estimate; no recalibration
chi	1.127	0.100	5.000	False	held at reference estimate; no recalibration
first birth fixed cost	0.507	0.000	8.000	False	held at reference estimate; no recalibration
kappa fert	0.209	0.020	50.000	True	held at reference estimate; no recalibration
kappa fert continuation	0.482	0.020	50.000	True	held at reference estimate; no recalibration
theta0	0.088	0.000	8.000	False	held at reference estimate; no recalibration
h P	1.890	0.100	2.300	False	held at reference estimate; no recalibration
theta1	0.008				externally fixed B15
psi child	0.134				held at reference; no fertility renormalization
payroll tax	0.080				derived from adopted pension ratio and baseline demographics
pension period	0.918				endogenous balanced PAYGO
housing supply elasticity	0.630				retained external setting
tenure choice kappa	0.005				retained external setting
alpha cons	0.733				retained external setting
sigma	2.000				retained external setting

Floor, linear benefit: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
selling cost	0.060				retained external setting
financed share	0.800				retained external setting
annual depreciation	0.014				author adopted input
period depreciation	0.055				four-year compounded
annual property tax	0.011				author adopted input
period property tax	0.042				four-year linear source convention
income process	15.000				retained B15 persistent-state count
entrant conversion factor	0.500				legacy child-departure diagnostic; inactive in split-birth entry
adult entry birth to household conversion	0.476				effective closed stationary birth conversion
child benefit exponent	1.000				fixed diagnostic restriction; not estimated
utility reference rent	0.110				fixed experimental common normalization
pension to gross worker earnings	0.229				adopted CPS2007 target
child benefit curvature	0.000				fixed trial value; distinct from fertility taste-shock scales
child benefit CRRA coefficient	0.134				ψ in $\psi \cdot m^{(1-\kappa)/(1-\kappa)}$; preserves one-child benefit
delta alpha jump	0.000				fixed first-child housing share increment; no later-child increment
delta alpha	0.000				fixed at zero; no additional housing share loading for subsequent children

Floor, concave benefit: complete target fit

Same target definitions and weights across cases; normalization row is descriptive at fixed child benefit

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.100	2.042	-0.058		
cps_childlessness	0.198	0.192	-0.006	3.55e+04	1.373
cps_exactly_one	0.214	0.245	0.031	2.7e+04	26.515
nchs_mean_age	25.976	26.453	0.477	139.828	31.773
nchs_share30	0.249	0.254	0.005	1.39e+04	0.295
wealth_earnings	6.927	6.219	-0.708	7.595	3.803
bequest_wealth	0.007	0.007	-0.000406	5.17e+06	0.850
old_dispersion	3.516	2.913	-0.603	10.616	3.862
mean_rooms	5.608	6.539	0.931	128.021	111.019
ownership_30_55	0.676	0.762	0.086	2.34e+03	17.218
first_birth_rooms	1.465	0.783	-0.682	137.565	64.003
family_rooms	0.385	0.229	-0.156	280.528	6.851
recent_parent_ownership	0.128	0.095	-0.033	2.71e+04	28.702

Floor, concave benefit: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
H0	8.646	0.200	80.000	False	held at reference estimate; no recalibration
beta annual	0.962	0.940	0.990	False	held at reference estimate; no recalibration
chi	1.127	0.100	5.000	False	held at reference estimate; no recalibration
first birth fixed cost	0.507	0.000	8.000	False	held at reference estimate; no recalibration
kappa fert	0.209	0.020	50.000	True	held at reference estimate; no recalibration
kappa fert continuation	0.482	0.020	50.000	True	held at reference estimate; no recalibration
theta0	0.088	0.000	8.000	False	held at reference estimate; no recalibration
h P	1.890	0.100	2.300	False	held at reference estimate; no recalibration
theta1	0.008				externally fixed B15
psi child	0.134				held at reference; no fertility renormalization
payroll tax	0.080				derived from adopted pension ratio and baseline demographics
pension period	0.918				endogenous balanced PAYGO
housing supply elasticity	0.630				retained external setting
tenure choice kappa	0.005				retained external setting
alpha cons	0.733				retained external setting
sigma	2.000				retained external setting

Floor, concave benefit: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
selling cost	0.060				retained external setting
financed share	0.800				retained external setting
annual depreciation	0.014				author adopted input
period depreciation	0.055				four-year compounded
annual property tax	0.011				author adopted input
period property tax	0.042				four-year linear source convention
income process	15.000				retained B15 persistent-state count
entrant conversion factor	0.500				legacy child-departure diagnostic; inactive in split-birth entry
adult entry birth to household conversion	0.476				effective closed stationary birth conversion
child benefit exponent	0.860				fixed diagnostic restriction; not estimated
utility reference rent	0.110				fixed experimental common normalization
pension to gross worker earnings	0.229				adopted CPS2007 target
child benefit curvature	0.140				fixed trial value; distinct from fertility taste-shock scales
child benefit CRRA coefficient	0.115				ψ in $\psi \cdot m^{(1-\kappa)/(1-\kappa)}$; preserves one-child benefit
delta alpha jump	0.000				fixed first-child housing share increment; no later-child increment
delta alpha	0.000				fixed at zero; no additional housing share loading for subsequent children

No floor, no loading: complete target fit

Same target definitions and weights across cases; normalization row is descriptive at fixed child benefit

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.100	2.633	0.533		
cps_childlessness	0.198	0.033	-0.166	3.55e+04	974.513
cps_exactly_one	0.214	0.160	-0.053	2.7e+04	76.820
nchs_mean_age	25.976	23.781	-2.195	139.828	673.959
nchs_share30	0.249	0.108	-0.142	1.39e+04	278.104
wealth_earnings	6.927	6.071	-0.855	7.595	5.557
bequest_wealth	0.007	0.007	-0.00026	5.17e+06	0.350
old_dispersion	3.516	2.911	-0.605	10.616	3.882
mean_rooms	5.608	6.370	0.762	128.021	74.388
ownership_30_55	0.676	0.763	0.087	2.34e+03	17.779
first_birth_rooms	1.465	0.120	-1.345	137.565	249.013
family_rooms	0.385	0.261	-0.124	280.528	4.303
recent_parent_ownership	0.128	-0.096	-0.223	2.71e+04	1.35e+03

No floor, no loading: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
H0	8.646	0.200	80.000	False	held at reference estimate; no recalibration
beta annual	0.962	0.940	0.990	False	held at reference estimate; no recalibration
chi	1.127	0.100	5.000	False	held at reference estimate; no recalibration
first birth fixed cost	0.507	0.000	8.000	False	held at reference estimate; no recalibration
kappa fert	0.209	0.020	50.000	True	held at reference estimate; no recalibration
kappa fert continuation	0.482	0.020	50.000	True	held at reference estimate; no recalibration
theta0	0.088	0.000	8.000	False	held at reference estimate; no recalibration
h P	0.000				housing floor removed in this experiment
theta1	0.008				externally fixed B15
psi child	0.134				held at reference; no fertility renormalization
payroll tax	0.080				derived from adopted pension ratio and baseline demographics
pension period	0.918				endogenous balanced PAYGO
housing supply elasticity	0.630				retained external setting
tenure choice kappa	0.005				retained external setting
alpha cons	0.733				retained external setting
sigma	2.000				retained external setting

No floor, no loading: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
selling cost	0.060				retained external setting
financed share	0.800				retained external setting
annual depreciation	0.014				author adopted input
period depreciation	0.055				four-year compounded
annual property tax	0.011				author adopted input
period property tax	0.042				four-year linear source convention
income process	15.000				retained B15 persistent-state count
entrant conversion factor	0.500				legacy child-departure diagnostic; inactive in split-birth entry
adult entry birth to household conversion	0.476				effective closed stationary birth conversion
child benefit exponent	0.860				fixed diagnostic restriction; not estimated
utility reference rent	0.110				fixed experimental common normalization
pension to gross worker earnings	0.229				adopted CPS2007 target
child benefit curvature	0.140				fixed trial value; distinct from fertility taste-shock scales
child benefit CRRA coefficient	0.115				ψ in $\psi \cdot m^{(1-\kappa)/(1-\kappa)}$; preserves one-child benefit
delta alpha jump	0.000				fixed first-child housing share increment; no later-child increment
delta alpha	0.000				fixed at zero; no additional housing share loading for subsequent children

No floor, +10 points: complete target fit

Same target definitions and weights across cases; normalization row is descriptive at fixed child benefit

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.100	2.605	0.505		
cps_childlessness	0.198	0.037	-0.161	3.55e+04	920.025
cps_exactly_one	0.214	0.166	-0.048	2.7e+04	61.495
nchs_mean_age	25.976	23.910	-2.066	139.828	597.065
nchs_share30	0.249	0.114	-0.135	1.39e+04	253.131
wealth_earnings	6.927	6.236	-0.691	7.595	3.624
bequest_wealth	0.007	0.007	-0.000292	5.17e+06	0.440
old_dispersion	3.516	2.920	-0.596	10.616	3.775
mean_rooms	5.608	6.629	1.021	128.021	133.533
ownership_30_55	0.676	0.768	0.091	2.34e+03	19.530
first_birth_rooms	1.465	0.941	-0.524	137.565	37.840
family_rooms	0.385	0.224	-0.161	280.528	7.254
recent_parent_ownership	0.128	-0.048	-0.176	2.71e+04	834.184

No floor, +10 points: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
H0	8.646	0.200	80.000	False	held at reference estimate; no recalibration
beta annual	0.962	0.940	0.990	False	held at reference estimate; no recalibration
chi	1.127	0.100	5.000	False	held at reference estimate; no recalibration
first birth fixed cost	0.507	0.000	8.000	False	held at reference estimate; no recalibration
kappa fert	0.209	0.020	50.000	True	held at reference estimate; no recalibration
kappa fert continuation	0.482	0.020	50.000	True	held at reference estimate; no recalibration
theta0	0.088	0.000	8.000	False	held at reference estimate; no recalibration
h P	0.000				housing floor removed in this experiment
theta1	0.008				externally fixed B15
psi child	0.134				held at reference; no fertility renormalization
payroll tax	0.080				derived from adopted pension ratio and baseline demographics
pension period	0.918				endogenous balanced PAYGO
housing supply elasticity	0.630				retained external setting
tenure choice kappa	0.005				retained external setting
alpha cons	0.733				retained external setting
sigma	2.000				retained external setting

No floor, +10 points: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
selling cost	0.060				retained external setting
financed share	0.800				retained external setting
annual depreciation	0.014				author adopted input
period depreciation	0.055				four-year compounded
annual property tax	0.011				author adopted input
period property tax	0.042				four-year linear source convention
income process	15.000				retained B15 persistent-state count
entrant conversion factor	0.500				legacy child-departure diagnostic; inactive in split-birth entry
adult entry birth to household conversion	0.476				effective closed stationary birth conversion
child benefit exponent	0.860				fixed diagnostic restriction; not estimated
utility reference rent	0.110				fixed experimental common normalization
pension to gross worker earnings	0.229				adopted CPS2007 target
child benefit curvature	0.140				fixed trial value; distinct from fertility taste-shock scales
child benefit CRRA coefficient	0.115				psi in $\psi \cdot m^{(1-\kappa)/(1-\kappa)}$; preserves one-child benefit
delta alpha jump	0.100				fixed first-child housing share increment; no later-child increment
delta alpha	0.000				fixed at zero; no additional housing share loading for subsequent children

No floor, +20 points: complete target fit

Same target definitions and weights across cases; normalization row is descriptive at fixed child benefit

Moment	Target	Model	Gap	Weight	Loss contribution
initial_normalization	2.100	2.540	0.440		
cps_childlessness	0.198	0.050	-0.149	3.55e+04	784.224
cps_exactly_one	0.214	0.178	-0.035	2.7e+04	33.523
nchs_mean_age	25.976	24.210	-1.766	139.828	436.057
nchs_share30	0.249	0.130	-0.120	1.39e+04	198.822
wealth_earnings	6.927	6.427	-0.500	7.595	1.898
bequest_wealth	0.007	0.007	-0.000377	5.17e+06	0.734
old_dispersion	3.516	2.914	-0.602	10.616	3.851
mean_rooms	5.608	6.769	1.162	128.021	172.739
ownership_30_55	0.676	0.780	0.104	2.34e+03	25.284
first_birth_rooms	1.465	1.538	0.073	137.565	0.727
family_rooms	0.385	0.170	-0.215	280.528	13.006
recent_parent_ownership	0.128	0.004	-0.124	2.71e+04	413.146

No floor, +20 points: parameters and restrictions

All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
H0	8.646	0.200	80.000	False	held at reference estimate; no recalibration
beta annual	0.962	0.940	0.990	False	held at reference estimate; no recalibration
chi	1.127	0.100	5.000	False	held at reference estimate; no recalibration
first birth fixed cost	0.507	0.000	8.000	False	held at reference estimate; no recalibration
kappa fert	0.209	0.020	50.000	True	held at reference estimate; no recalibration
kappa fert continuation	0.482	0.020	50.000	True	held at reference estimate; no recalibration
theta0	0.088	0.000	8.000	False	held at reference estimate; no recalibration
h P	0.000				housing floor removed in this experiment
theta1	0.008				externally fixed B15
psi child	0.134				held at reference; no fertility renormalization
payroll tax	0.080				derived from adopted pension ratio and baseline demographics
pension period	0.918				endogenous balanced PAYGO
housing supply elasticity	0.630				retained external setting
tenure choice kappa	0.005				retained external setting
alpha cons	0.733				retained external setting
sigma	2.000				retained external setting

No floor, +20 points: parameters and restrictions

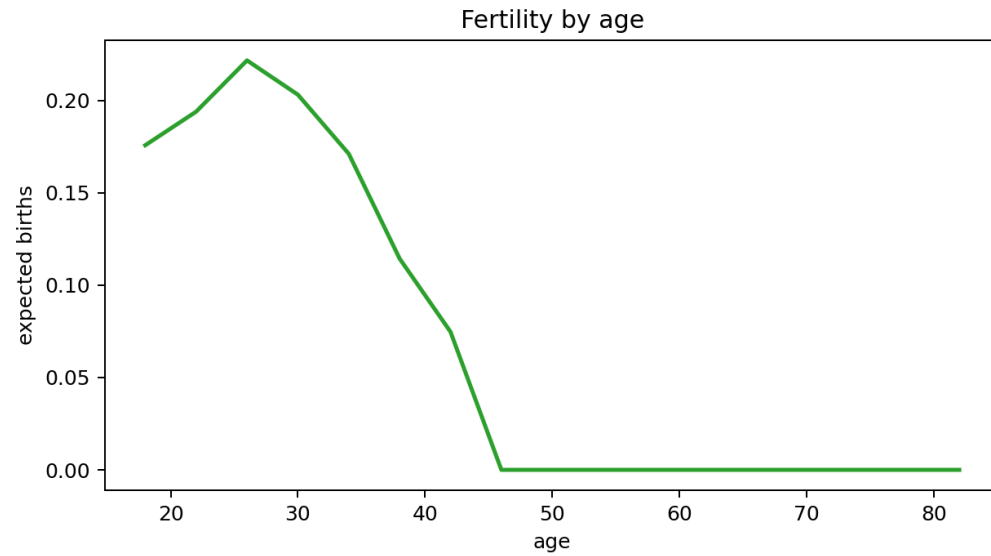
All values fixed in this diagnostic; inherited estimation bounds are shown where applicable

Parameter	Value	Lower	Upper	Near bound	Status
selling cost	0.060				retained external setting
financed share	0.800				retained external setting
annual depreciation	0.014				author adopted input
period depreciation	0.055				four-year compounded
annual property tax	0.011				author adopted input
period property tax	0.042				four-year linear source convention
income process	15.000				retained B15 persistent-state count
entrant conversion factor	0.500				legacy child-departure diagnostic; inactive in split-birth entry
adult entry birth to household conversion	0.476				effective closed stationary birth conversion
child benefit exponent	0.860				fixed diagnostic restriction; not estimated
utility reference rent	0.110				fixed experimental common normalization
pension to gross worker earnings	0.229				adopted CPS2007 target
child benefit curvature	0.140				fixed trial value; distinct from fertility taste-shock scales
child benefit CRRA coefficient	0.115				psi in $\psi \cdot m^{(1-\kappa)/(1-\kappa)}$; preserves one-child benefit
delta alpha jump	0.200				fixed first-child housing share increment; no later-child increment
delta alpha	0.000				fixed at zero; no additional housing share loading for subsequent children

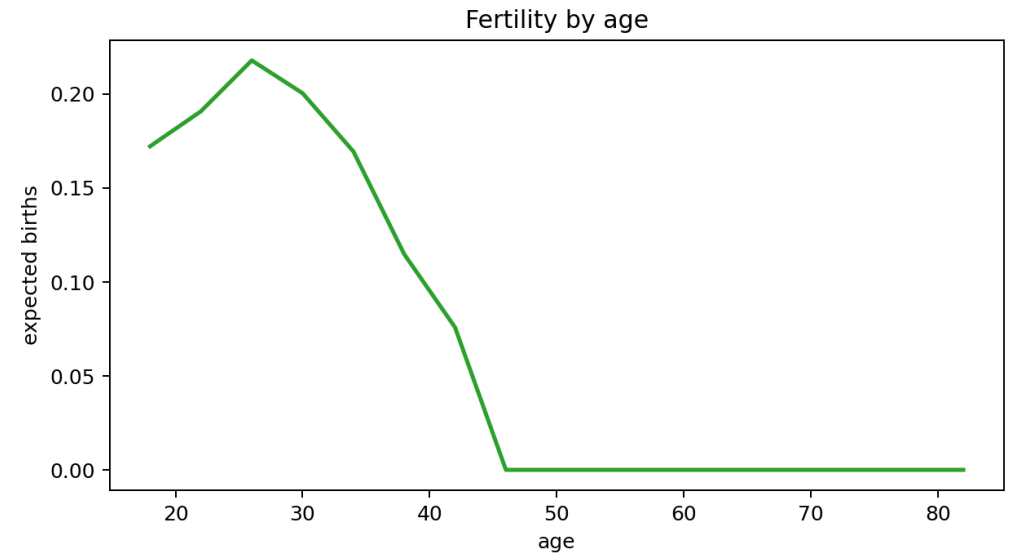
fertility by age

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



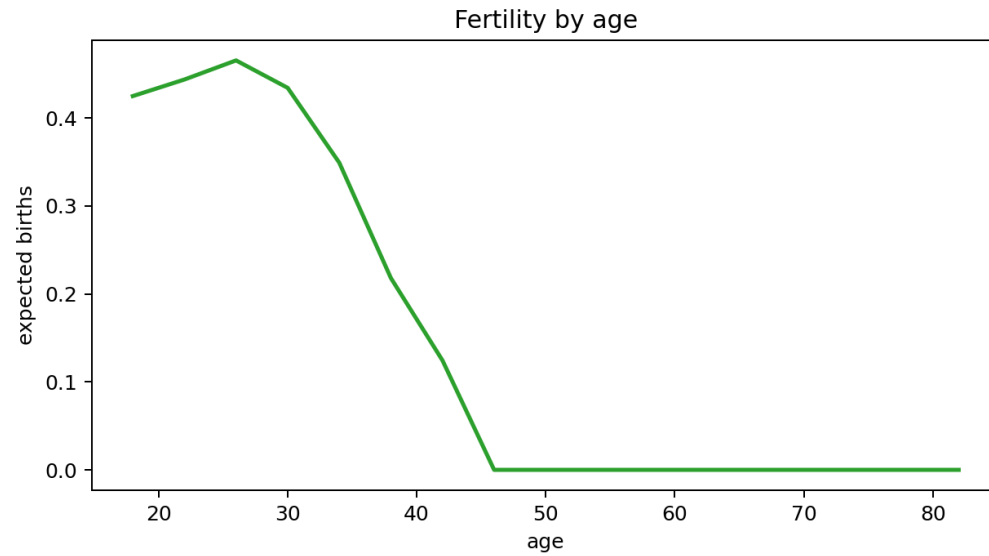
Floor, concave benefit



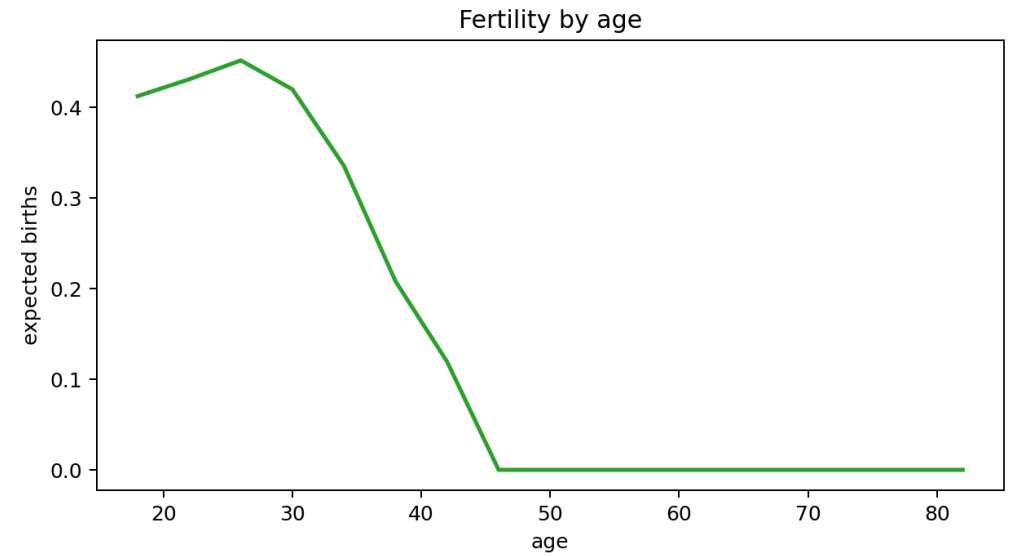
fertility by age

Standard diagnostic; same graph set for every completed case

No floor, no loading



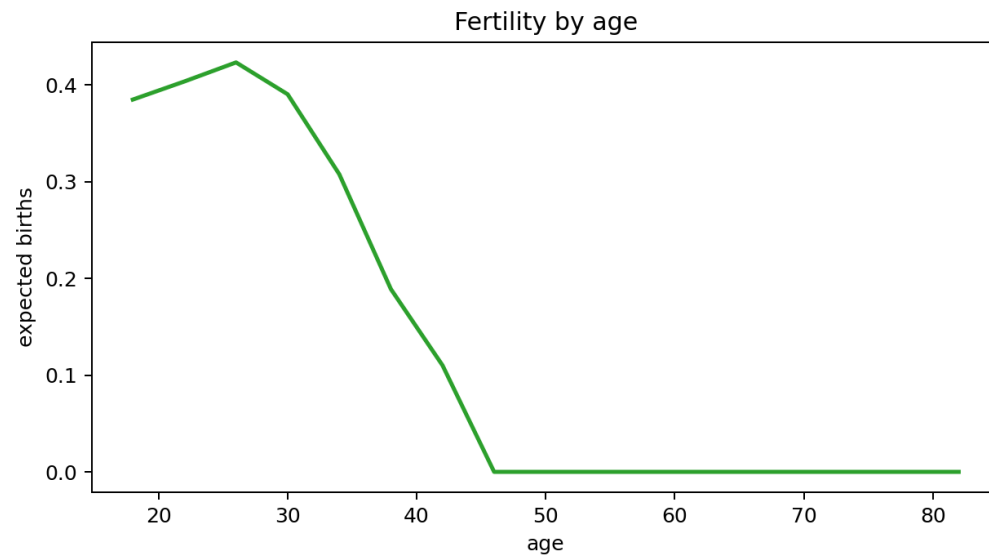
No floor, +10 points



fertility by age

Standard diagnostic; same graph set for every completed case

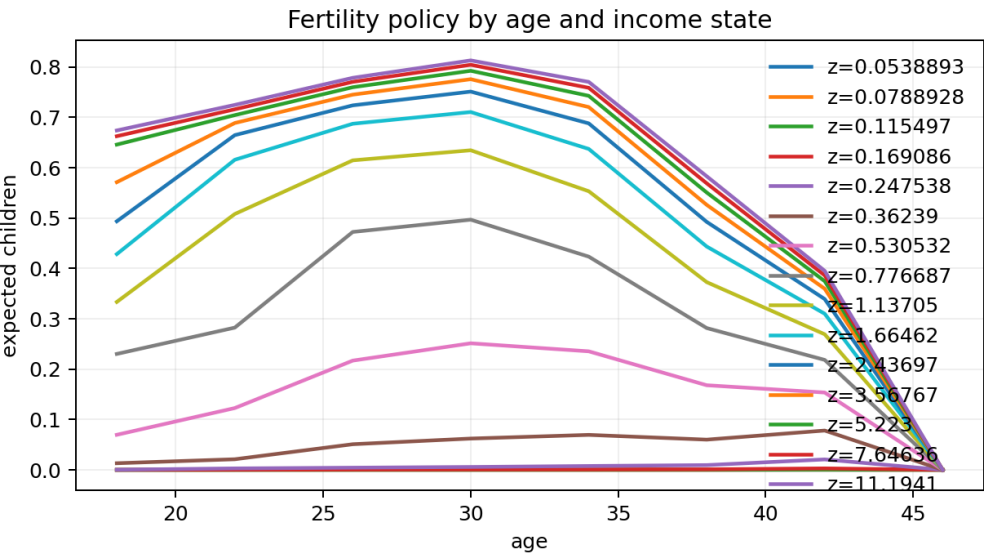
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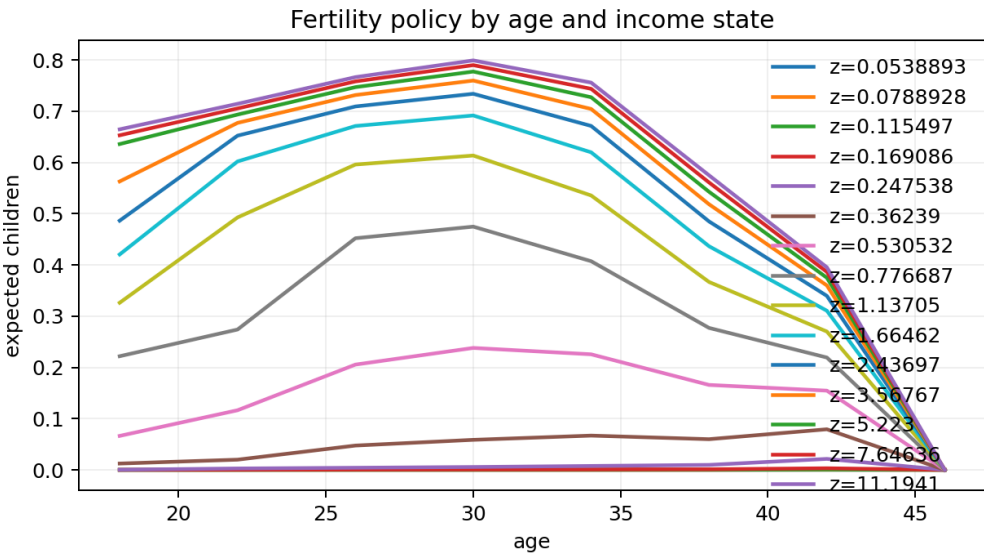
fertility policy by age income state

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit

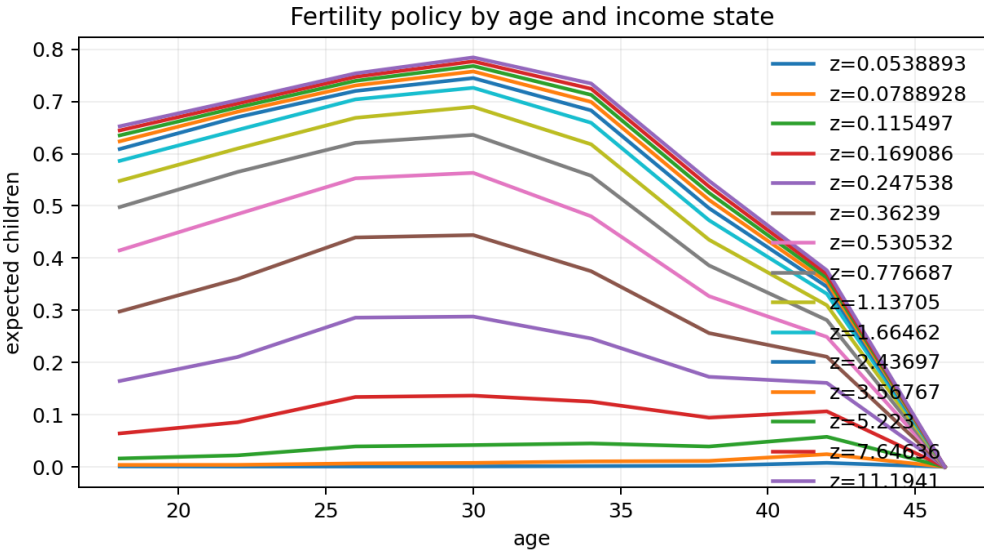
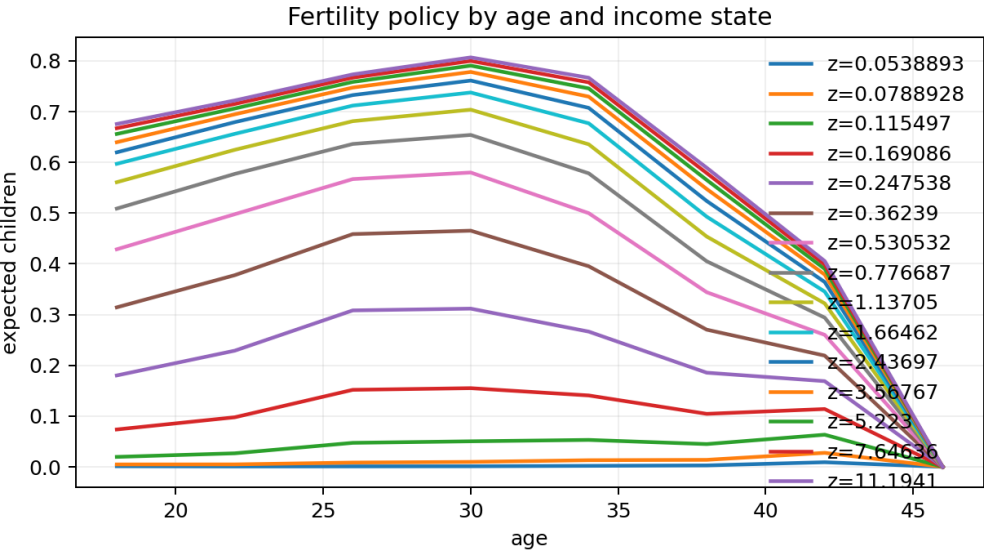


fertility policy by age income state

Standard diagnostic; same graph set for every completed case

No floor, no loading

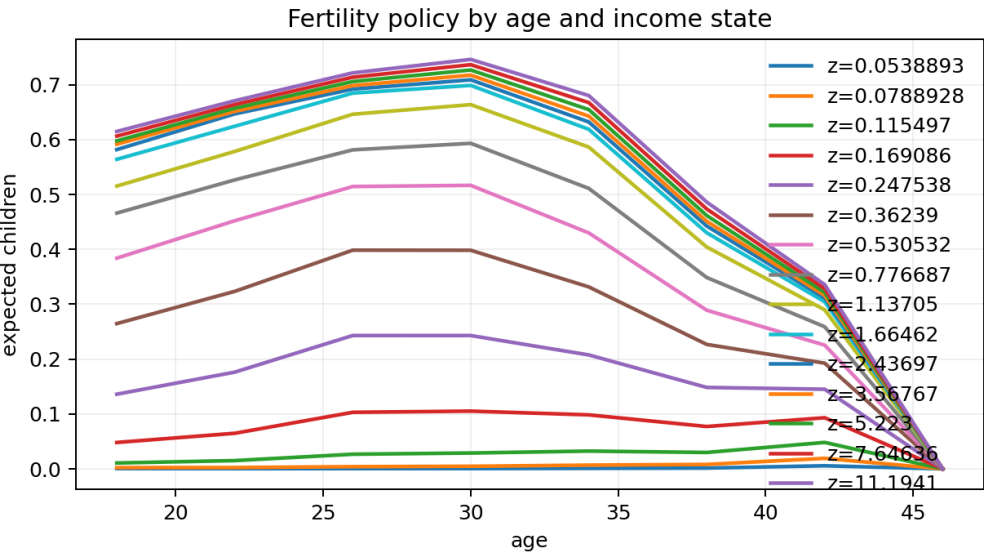
No floor, +10 points



fertility policy by age income state

Standard diagnostic; same graph set for every completed case

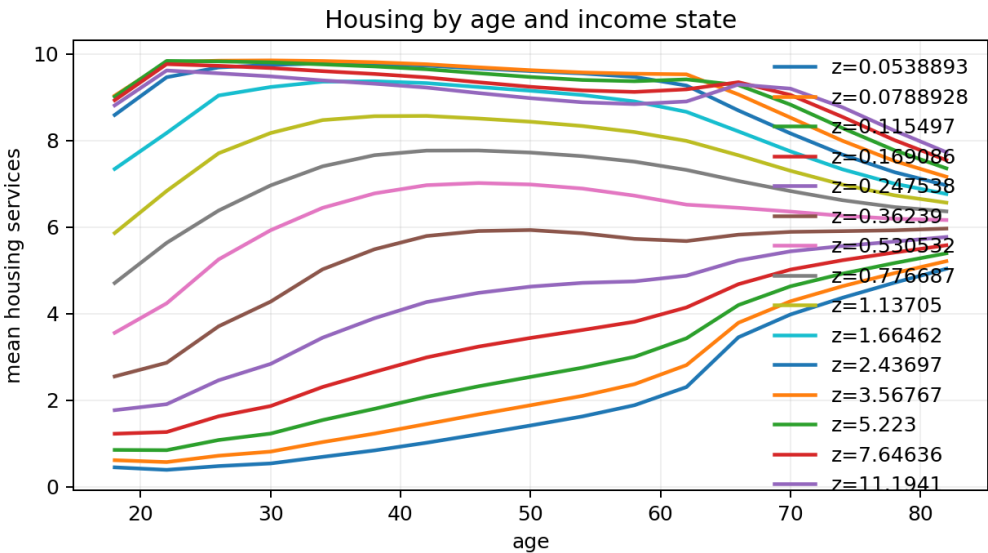
No floor, +20 points



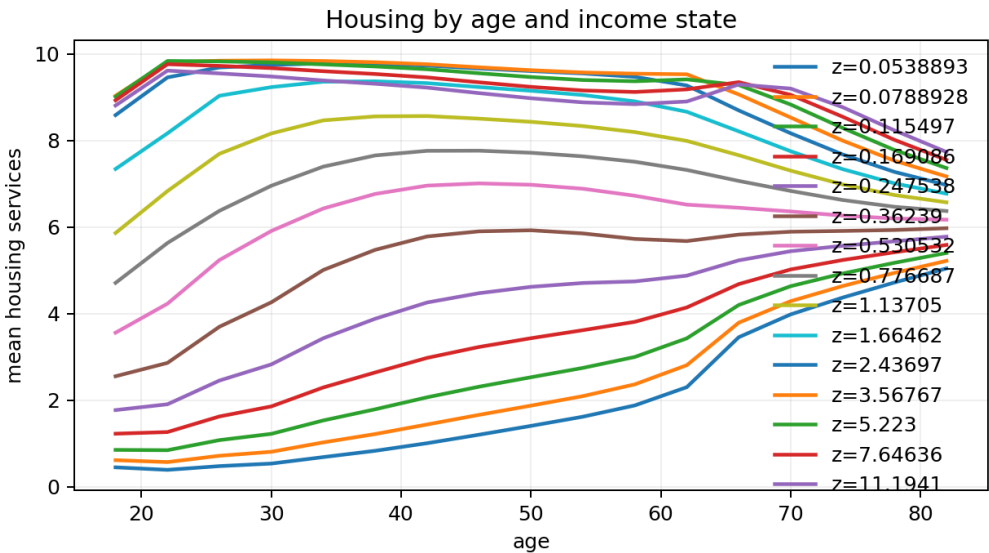
housing by age income state

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



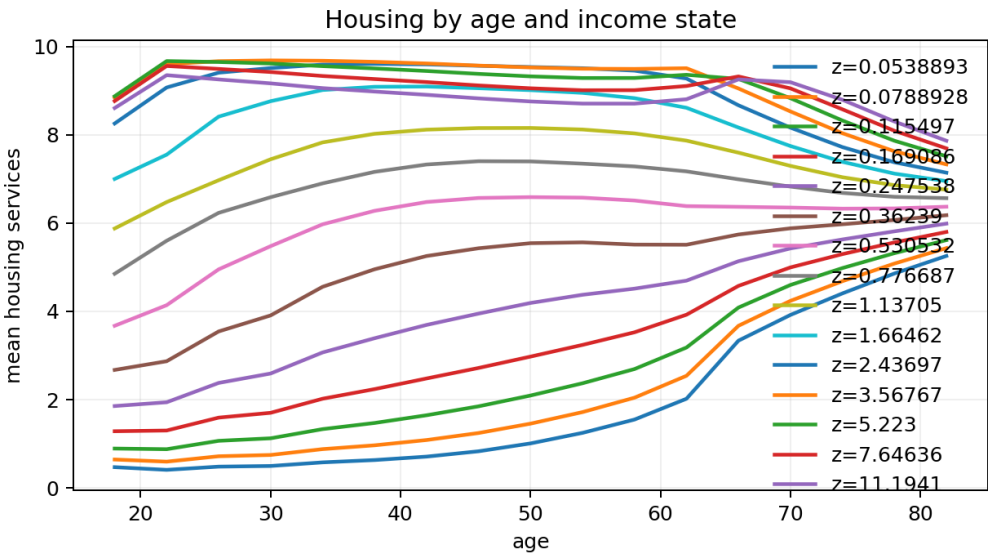
Floor, concave benefit



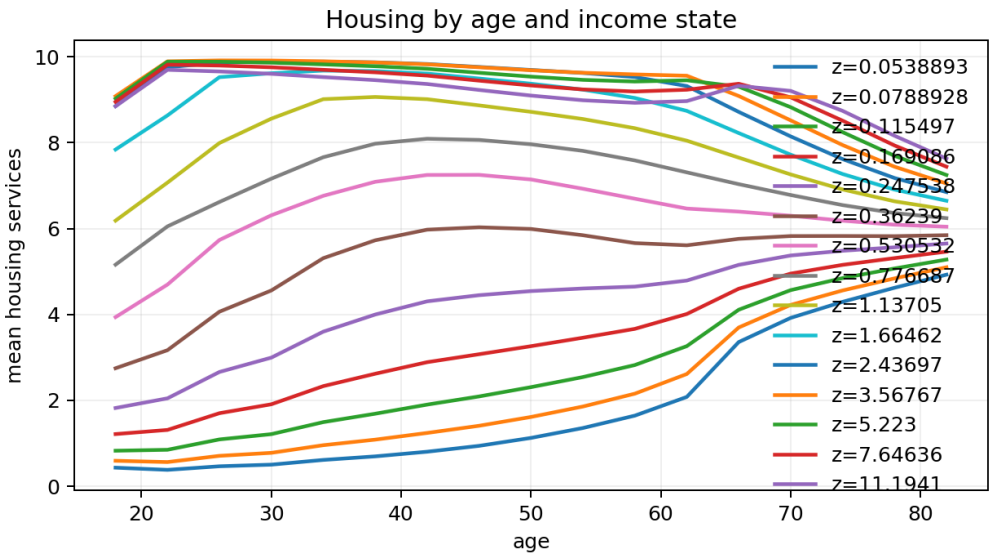
housing by age income state

Standard diagnostic; same graph set for every completed case

No floor, no loading



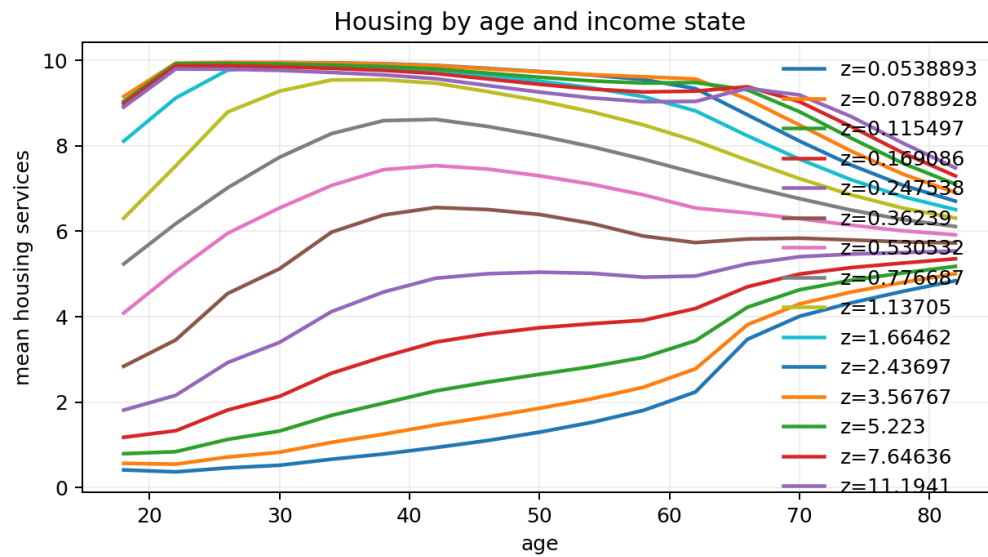
No floor, +10 points



housing by age income state

Standard diagnostic; same graph set for every completed case

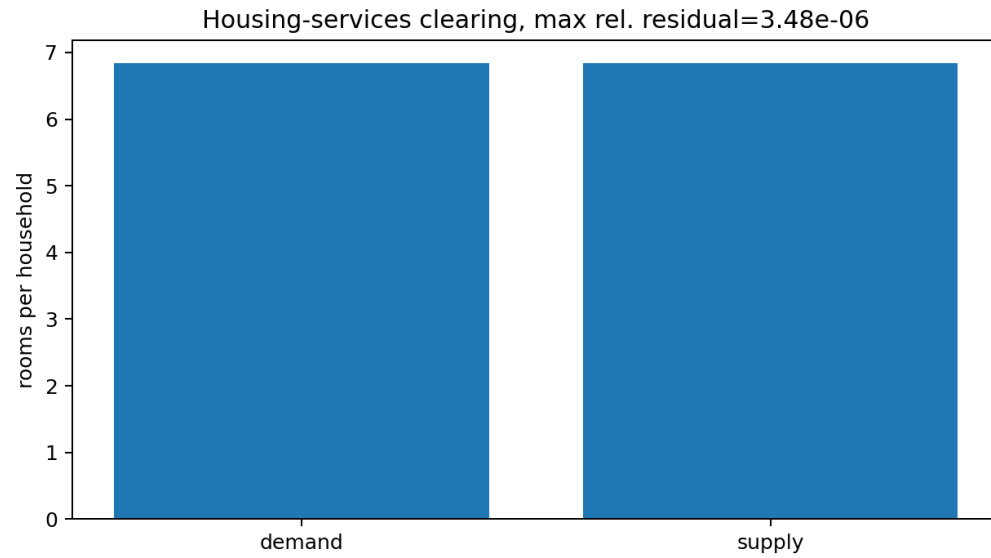
No floor, +20 points



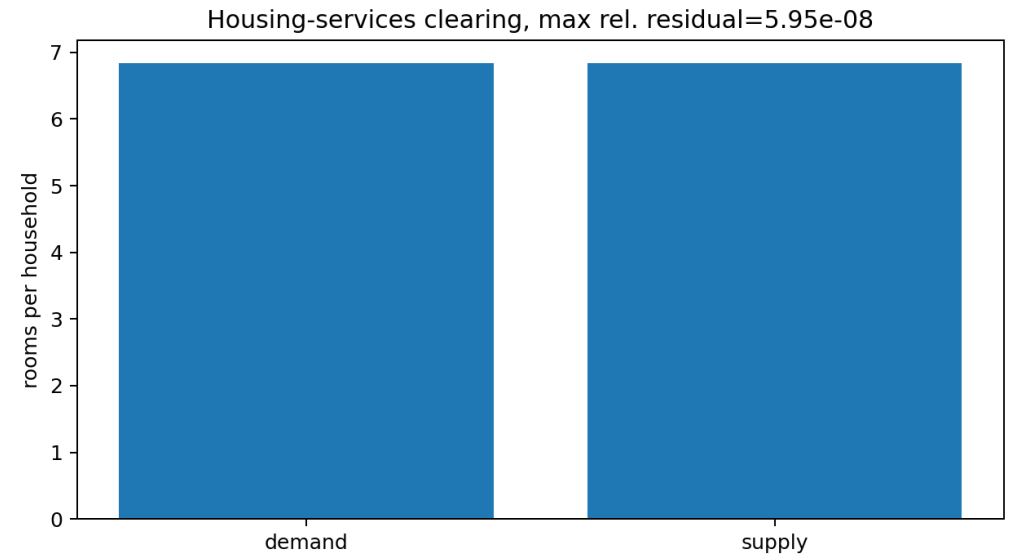
housing market

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit

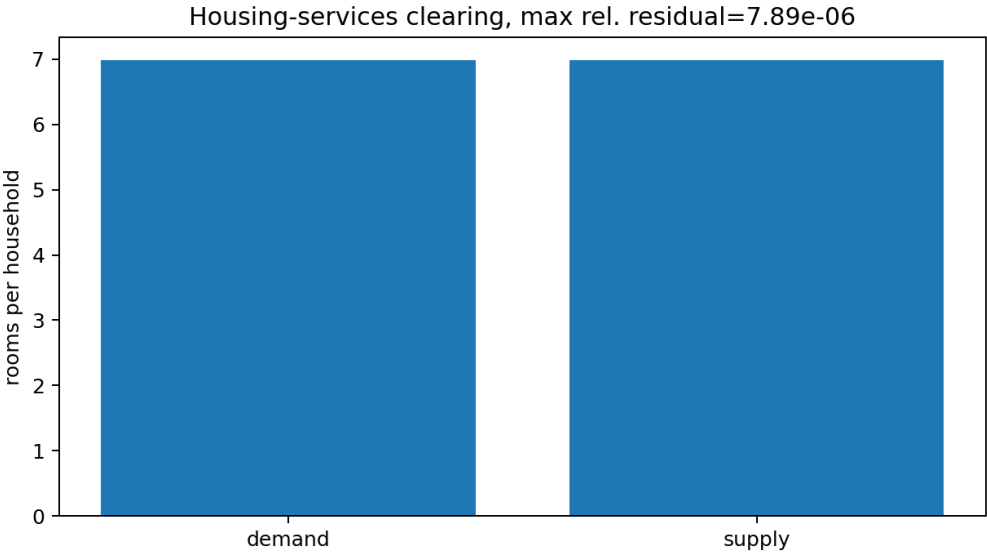
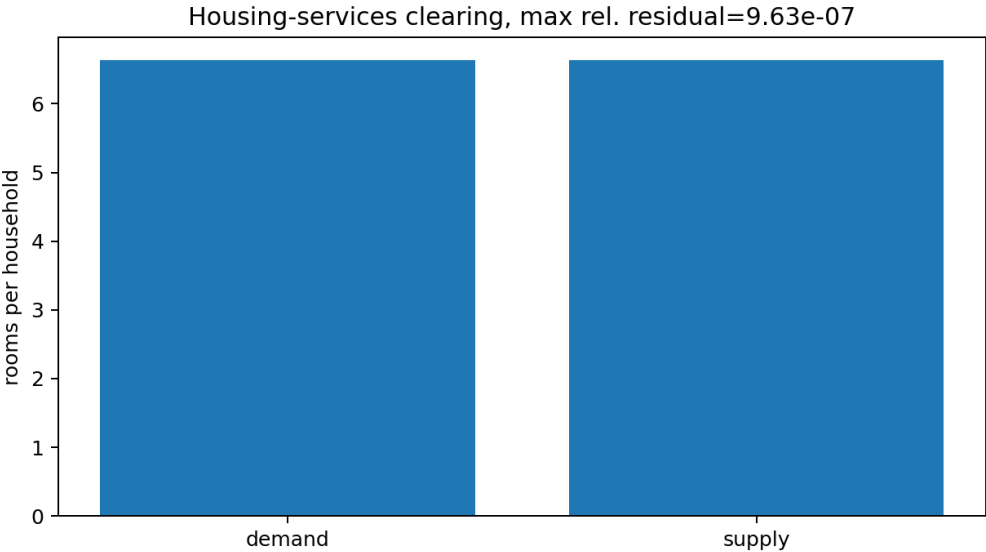


housing market

Standard diagnostic; same graph set for every completed case

No floor, no loading

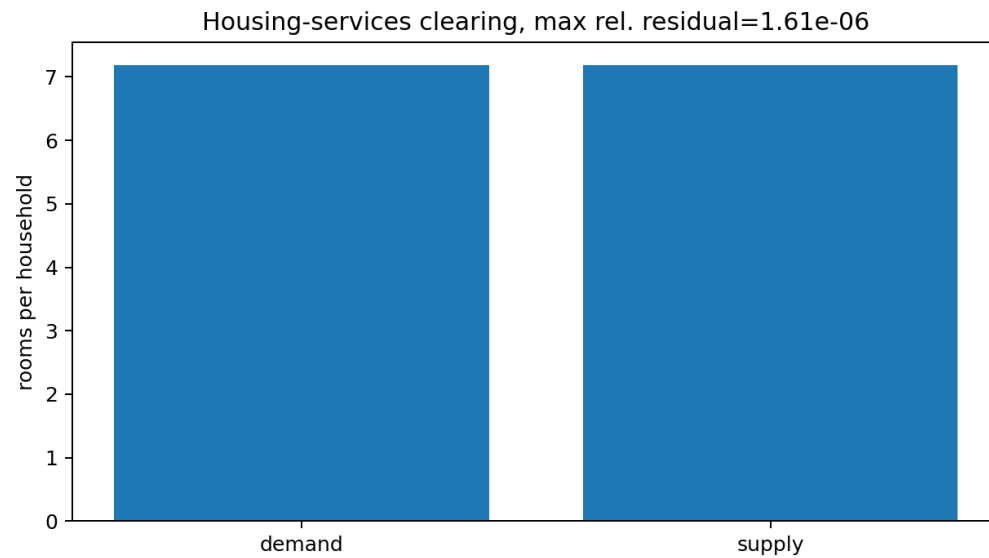
No floor, +10 points



housing market

Standard diagnostic; same graph set for every completed case

No floor, +20 points



housing prices

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit



housing prices

Standard diagnostic; same graph set for every completed case

No floor, no loading



No floor, +10 points



housing prices

Standard diagnostic; same graph set for every completed case

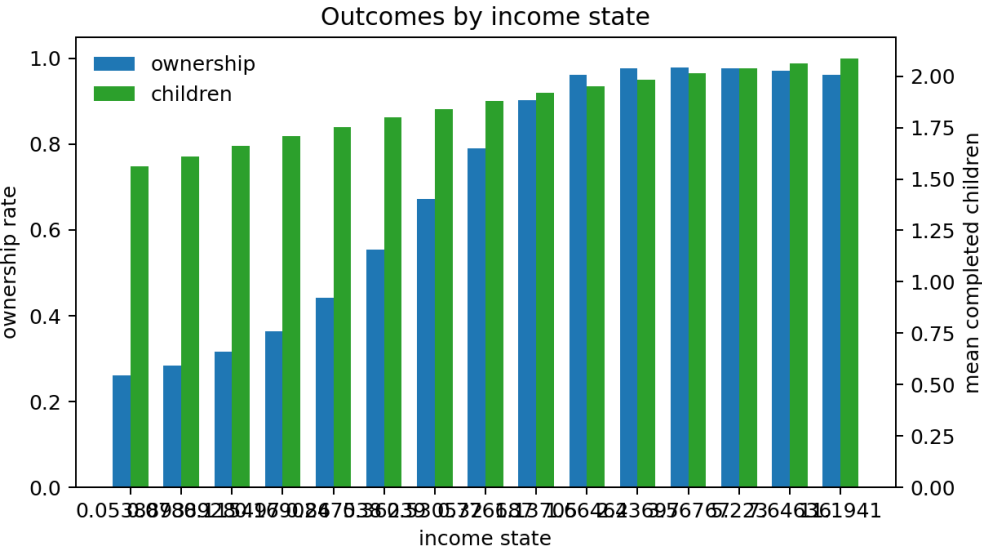
No floor, +20 points



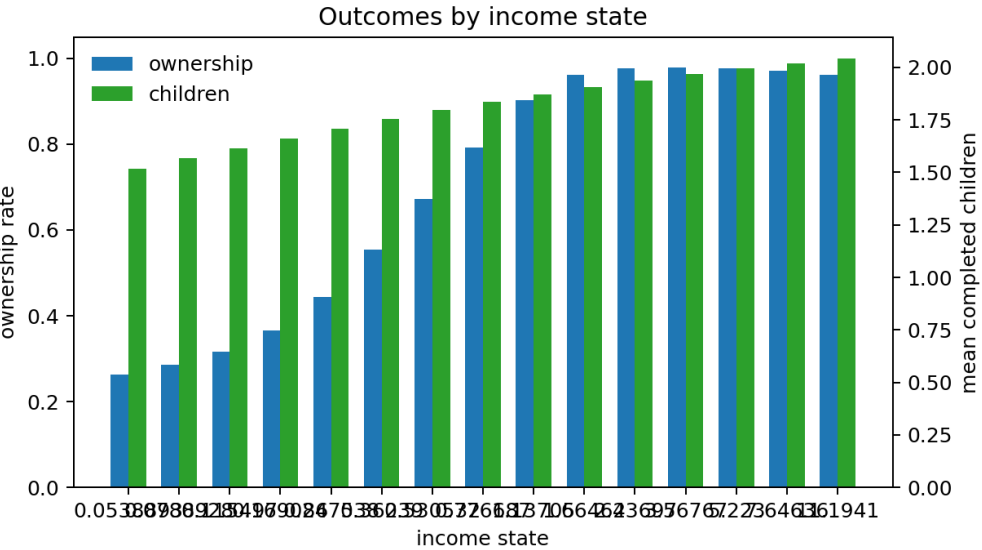
income state outcomes

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit

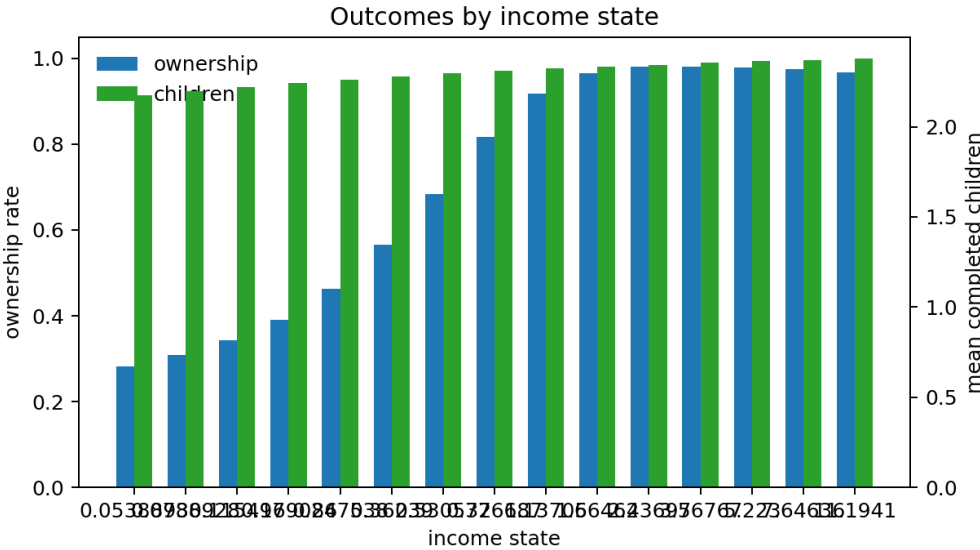
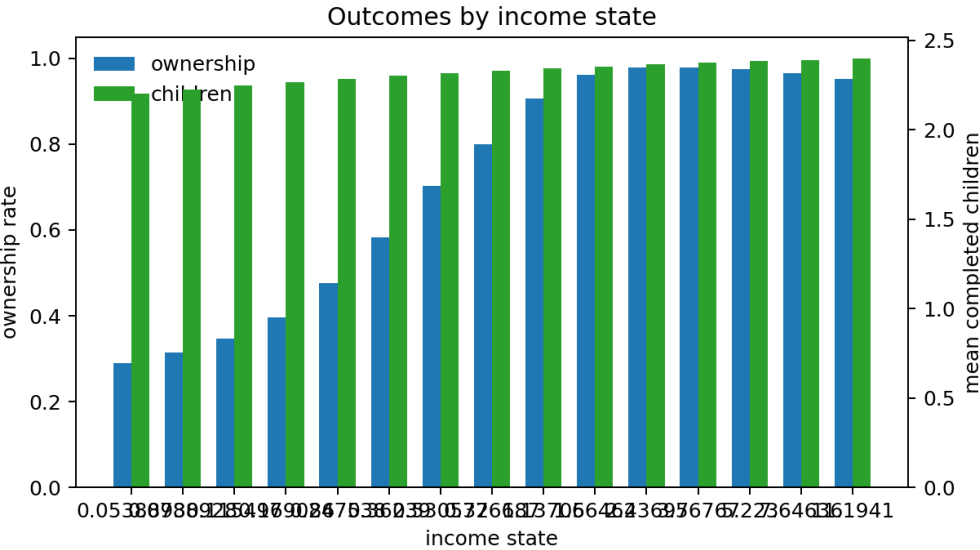


income state outcomes

Standard diagnostic; same graph set for every completed case

No floor, no loading

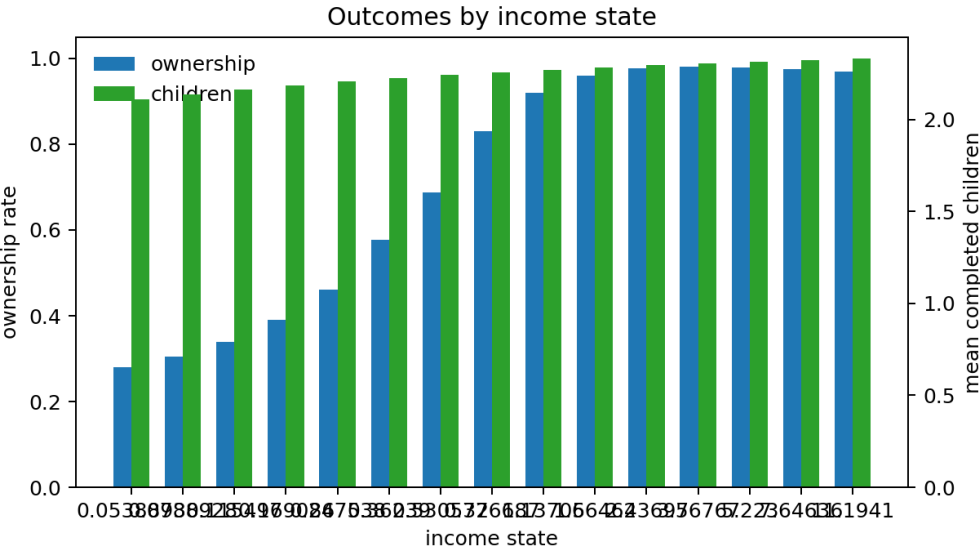
No floor, +10 points



income state outcomes

Standard diagnostic; same graph set for every completed case

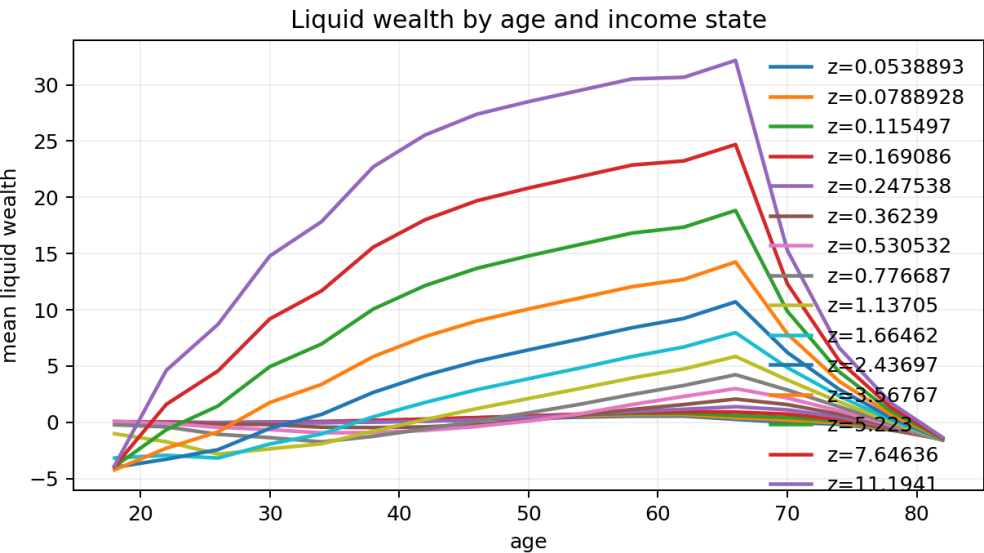
No floor, +20 points



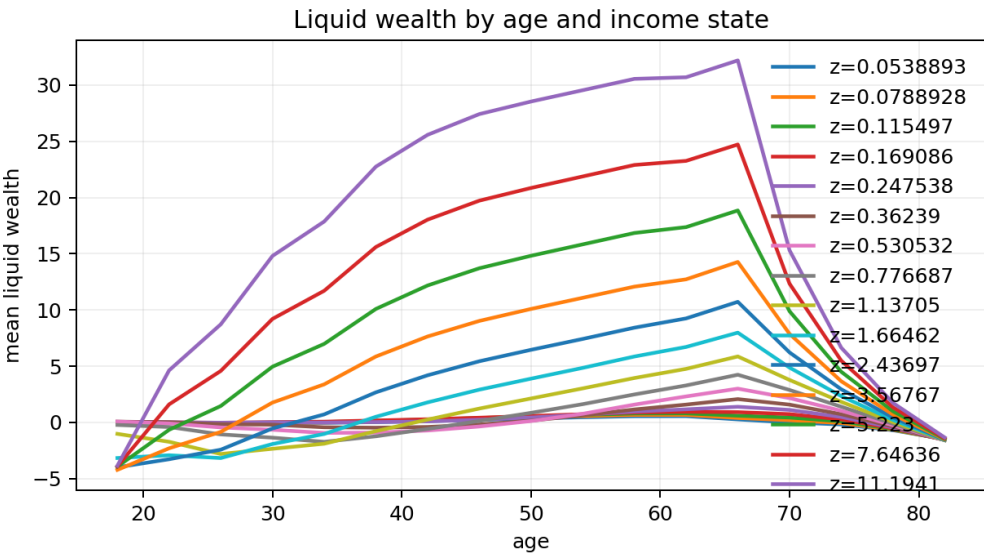
liquid wealth by age income state

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



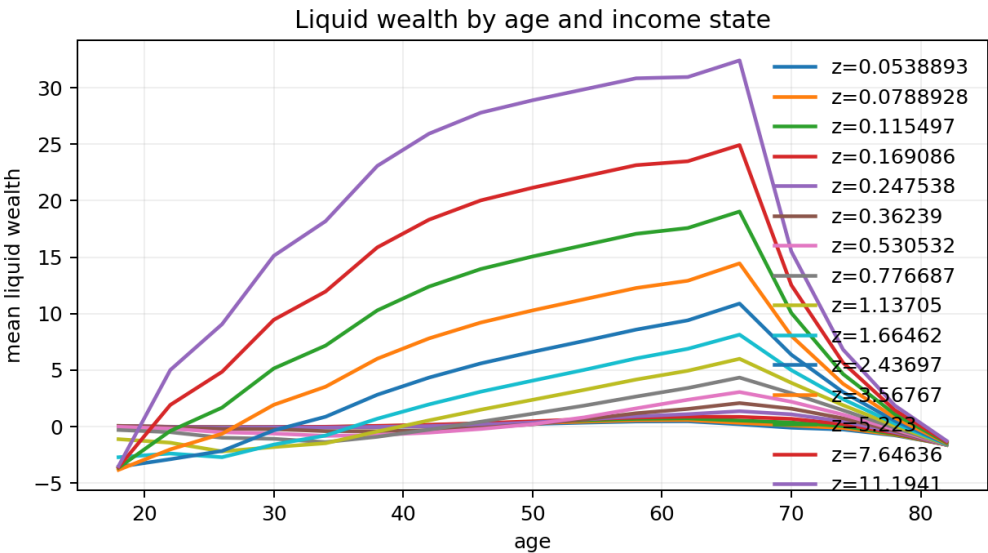
Floor, concave benefit



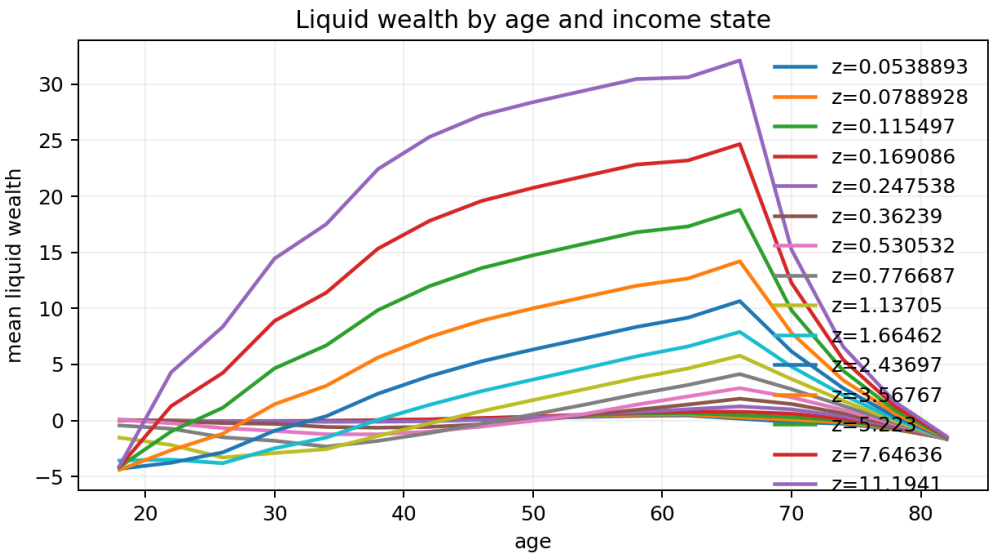
liquid wealth by age income state

Standard diagnostic; same graph set for every completed case

No floor, no loading



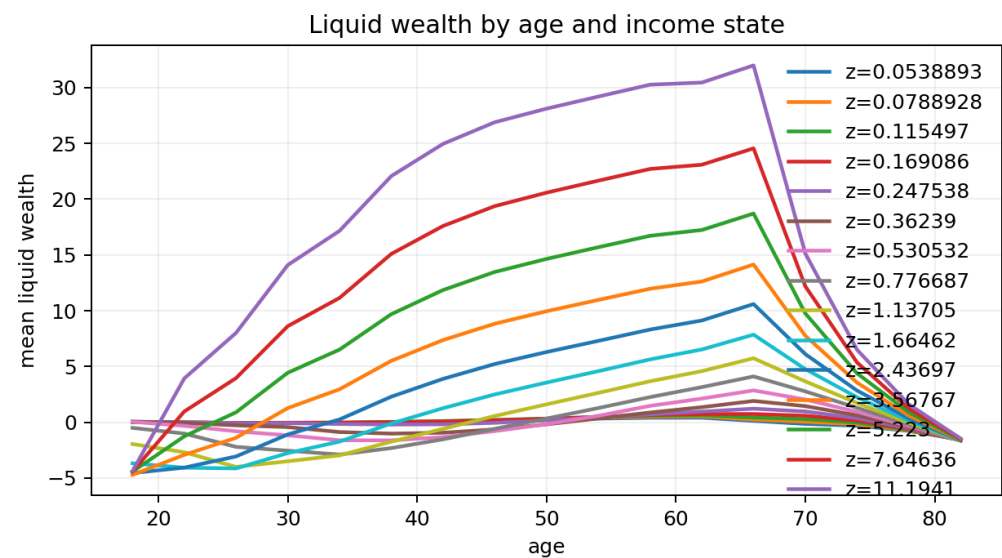
No floor, +10 points



liquid wealth by age income state

Standard diagnostic; same graph set for every completed case

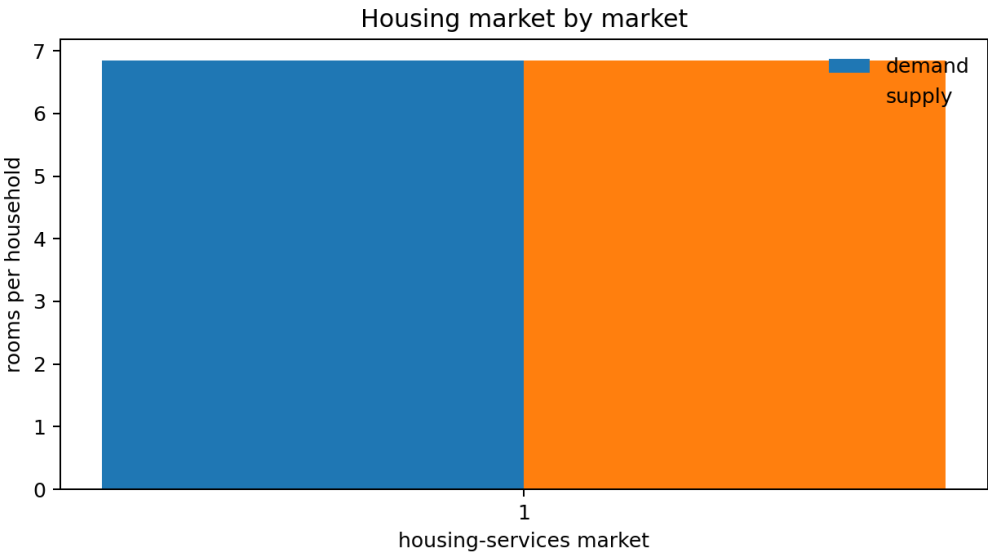
No floor, +20 points



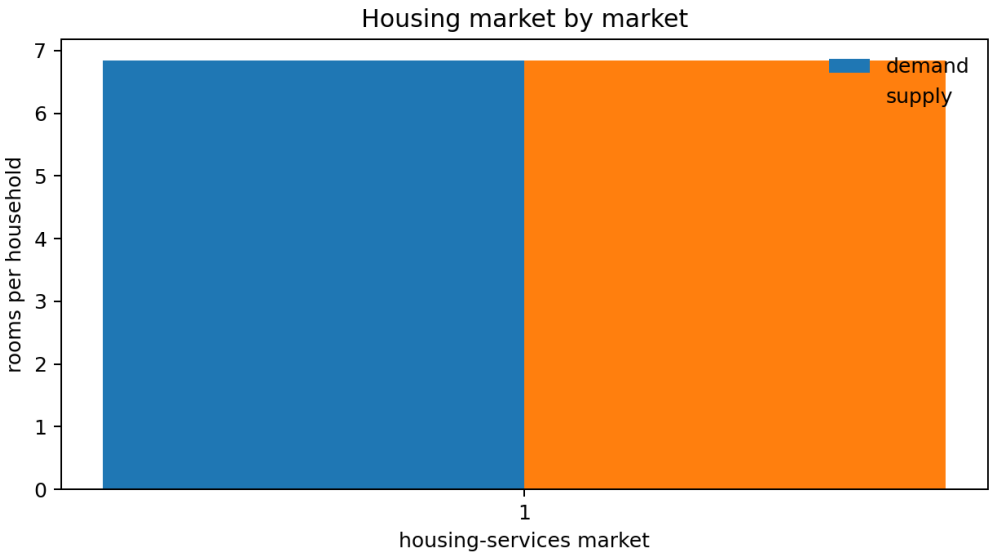
market clearing by market

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



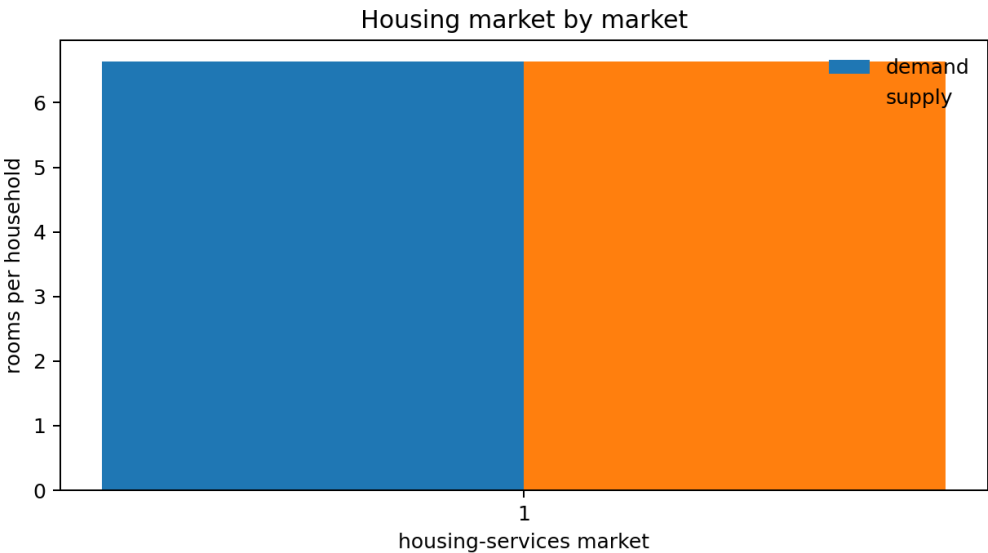
Floor, concave benefit



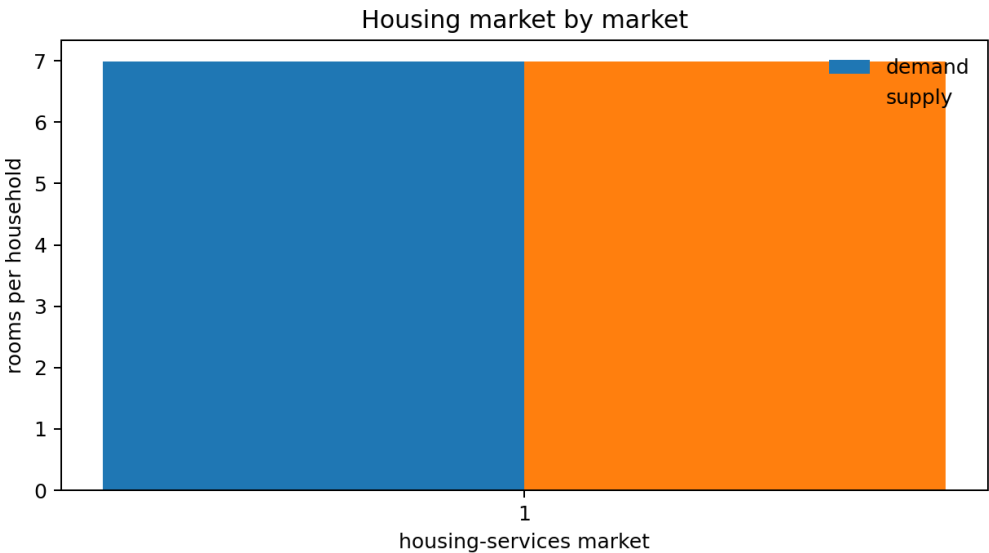
market clearing by market

Standard diagnostic; same graph set for every completed case

No floor, no loading



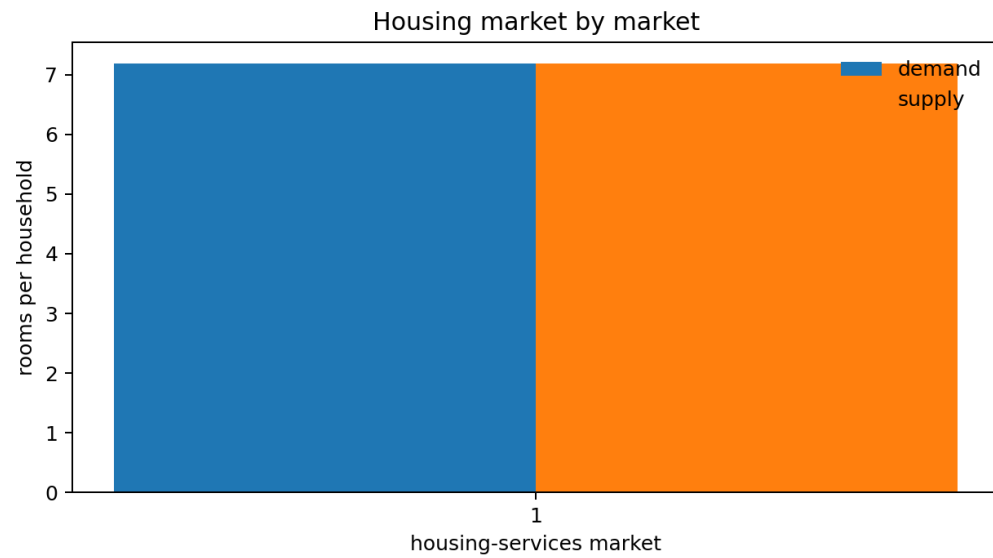
No floor, +10 points



market clearing by market

Standard diagnostic; same graph set for every completed case

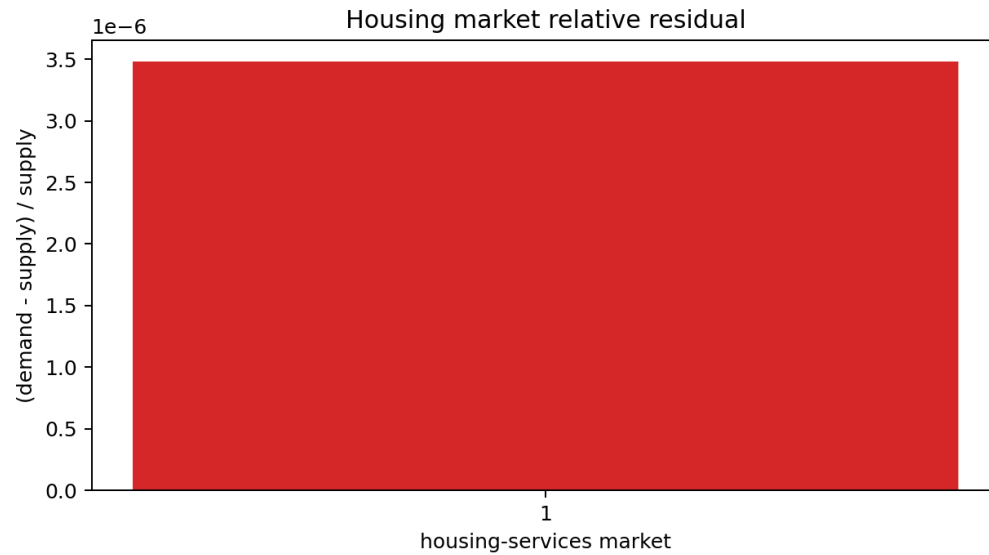
No floor, +20 points



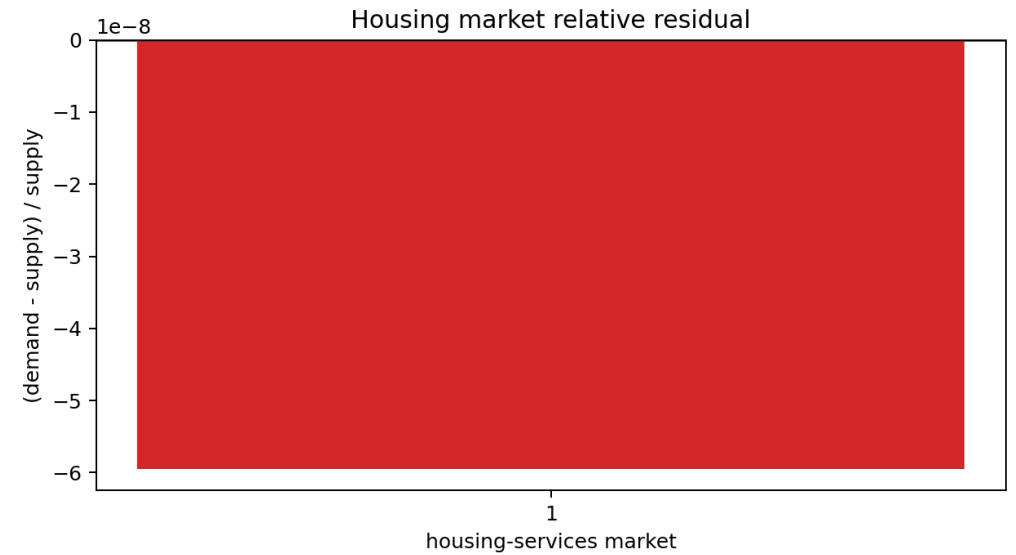
market clearing residuals

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



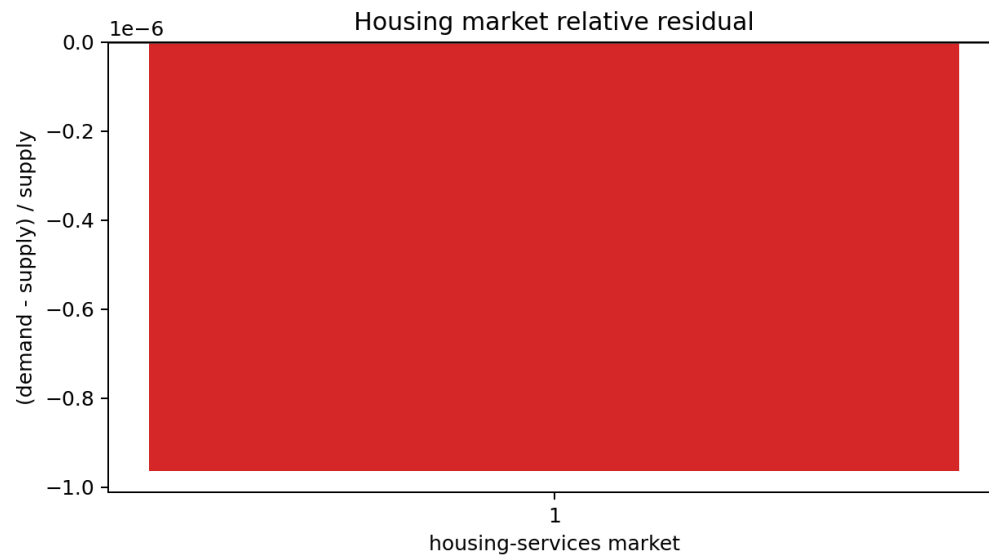
Floor, concave benefit



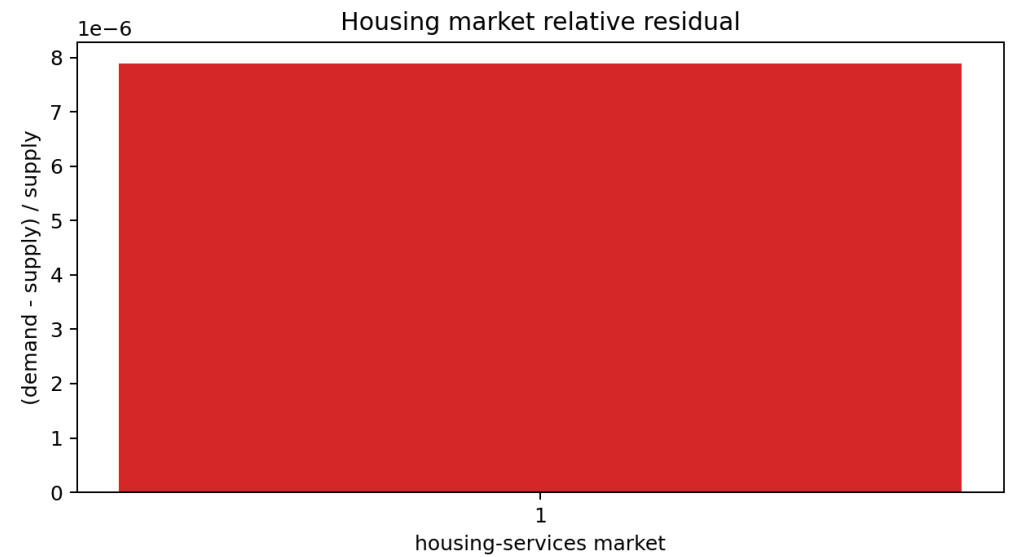
market clearing residuals

Standard diagnostic; same graph set for every completed case

No floor, no loading



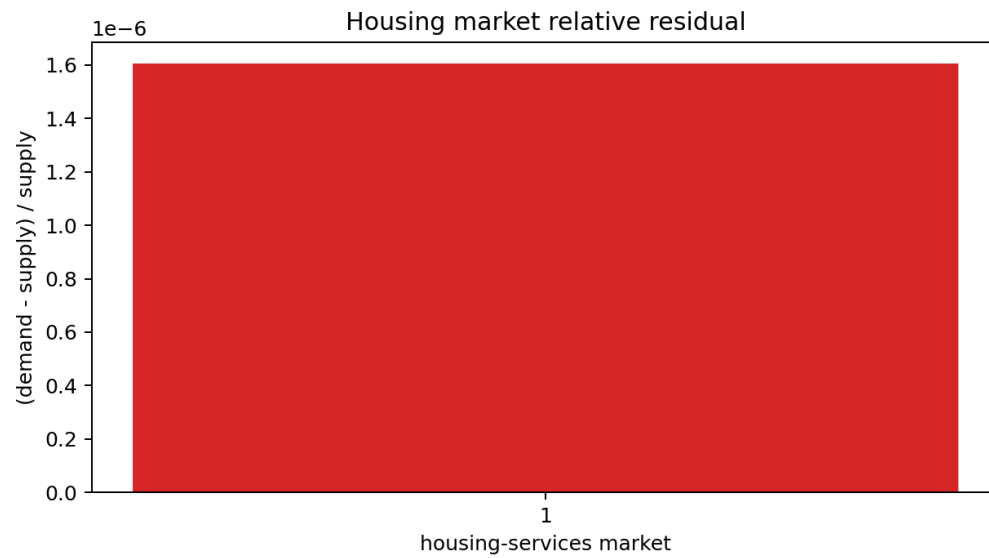
No floor, +10 points



market clearing residuals

Standard diagnostic; same graph set for every completed case

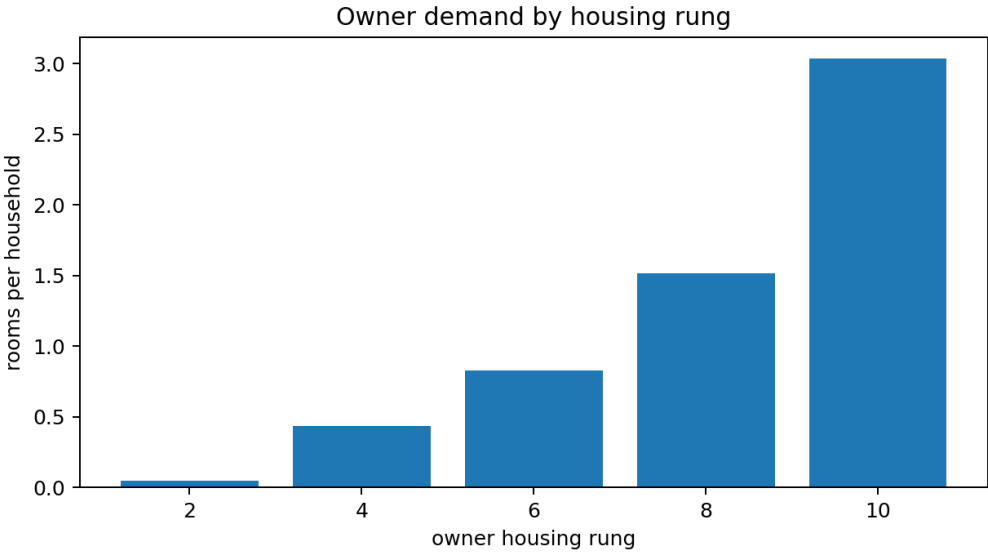
No floor, +20 points



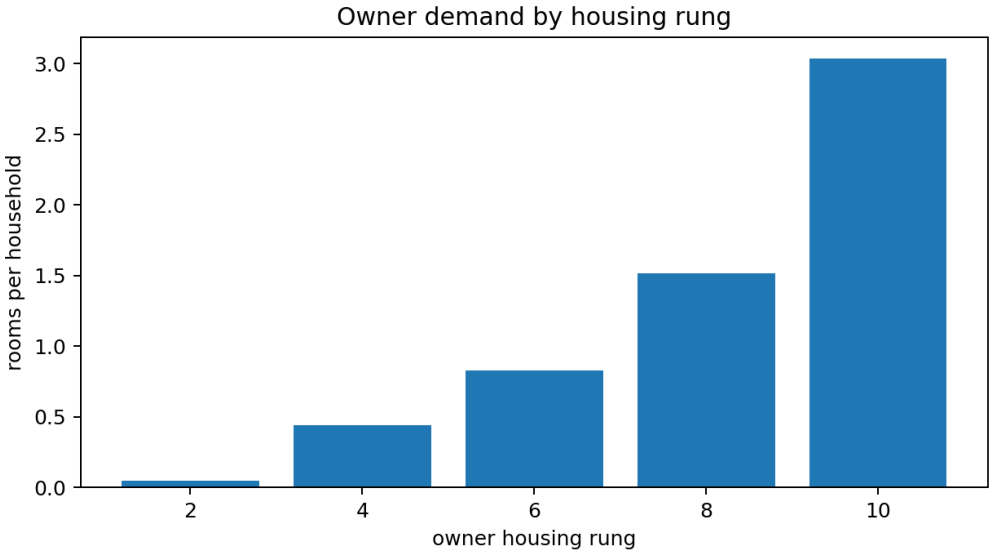
owner rungs

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit

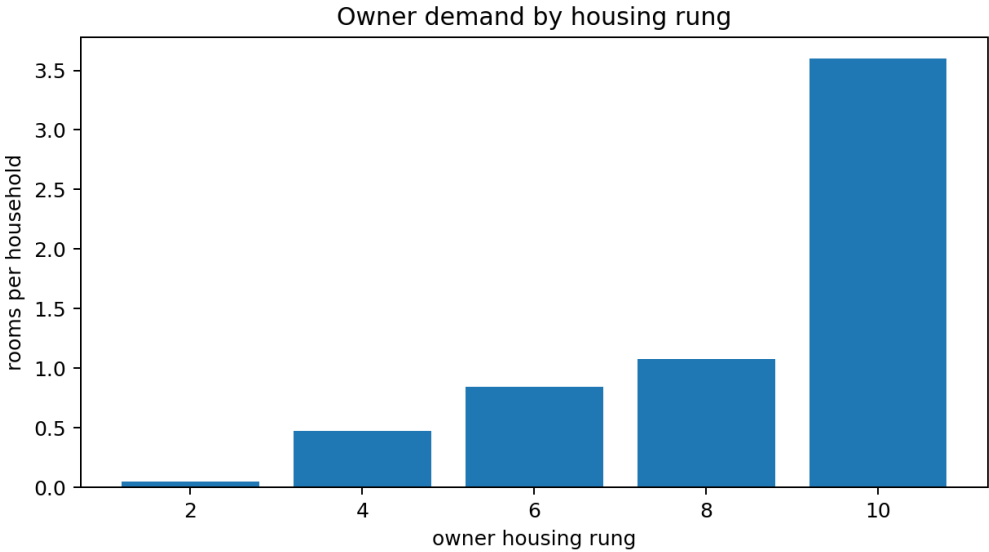
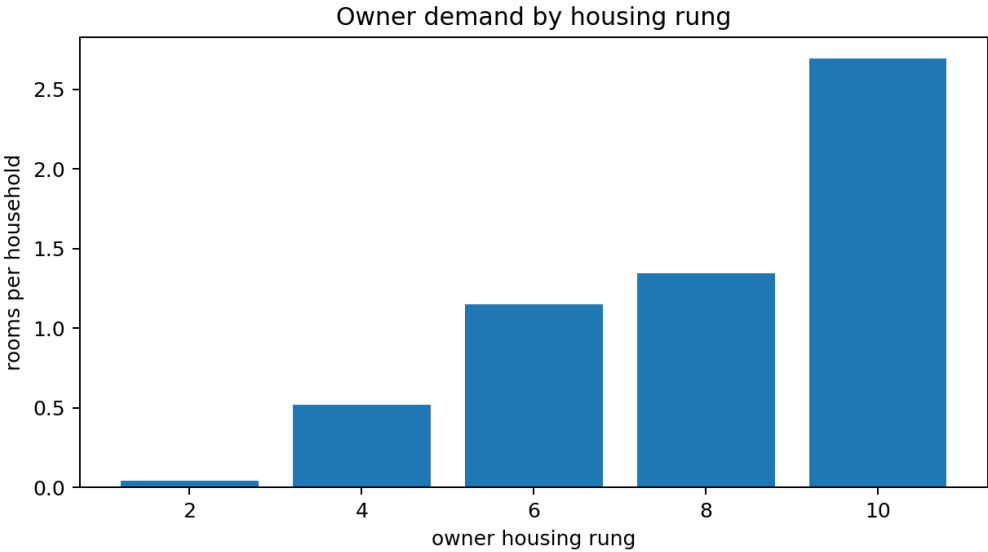


owner rungs

Standard diagnostic; same graph set for every completed case

No floor, no loading

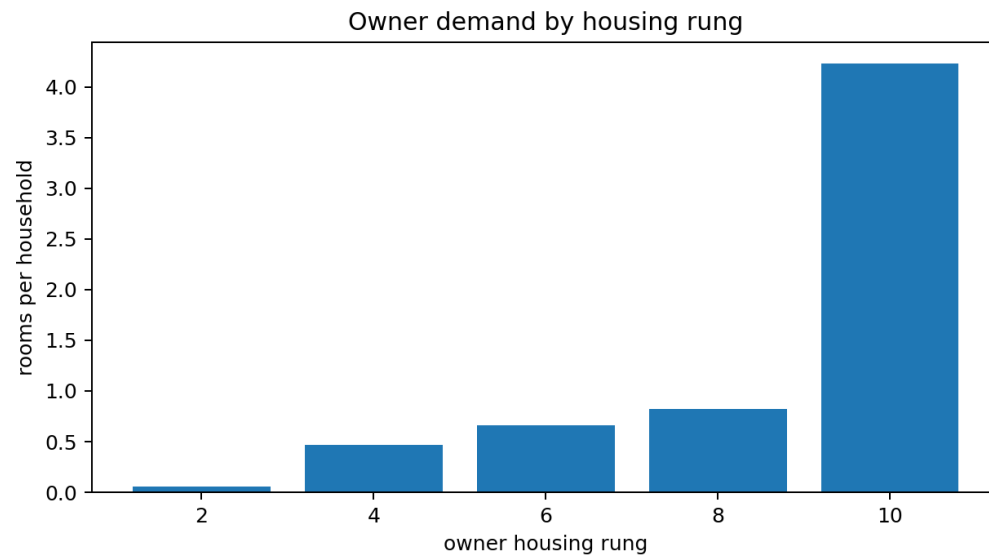
No floor, +10 points



owner rungs

Standard diagnostic; same graph set for every completed case

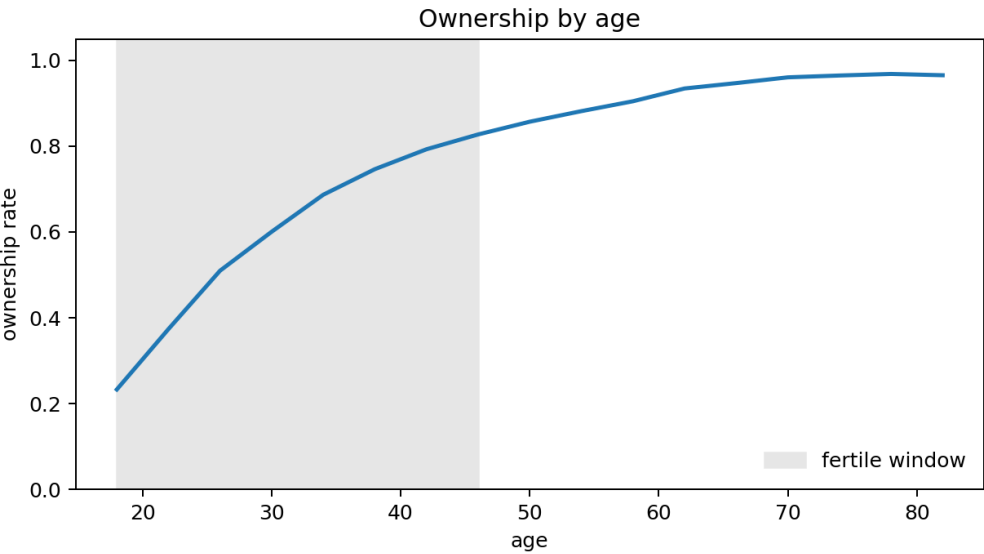
No floor, +20 points



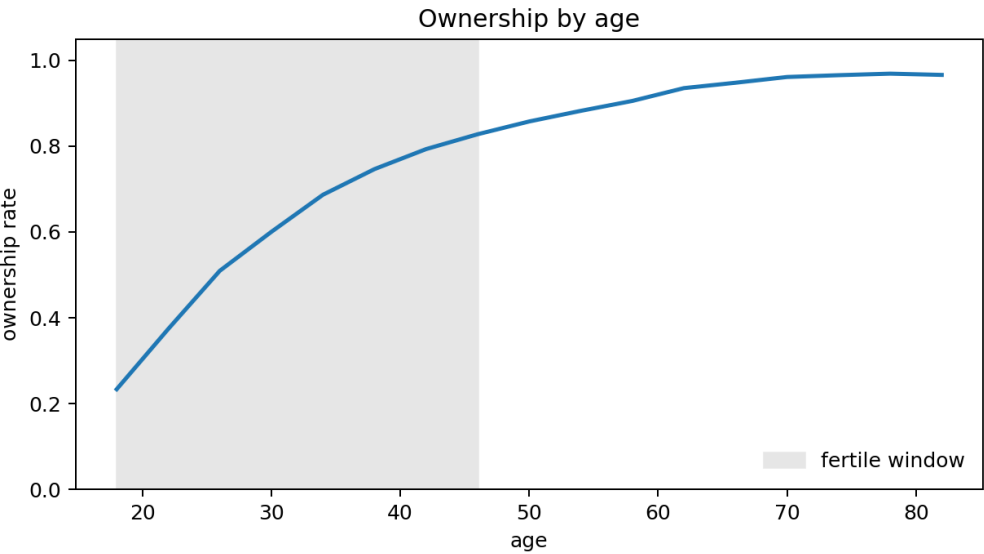
ownership by age

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



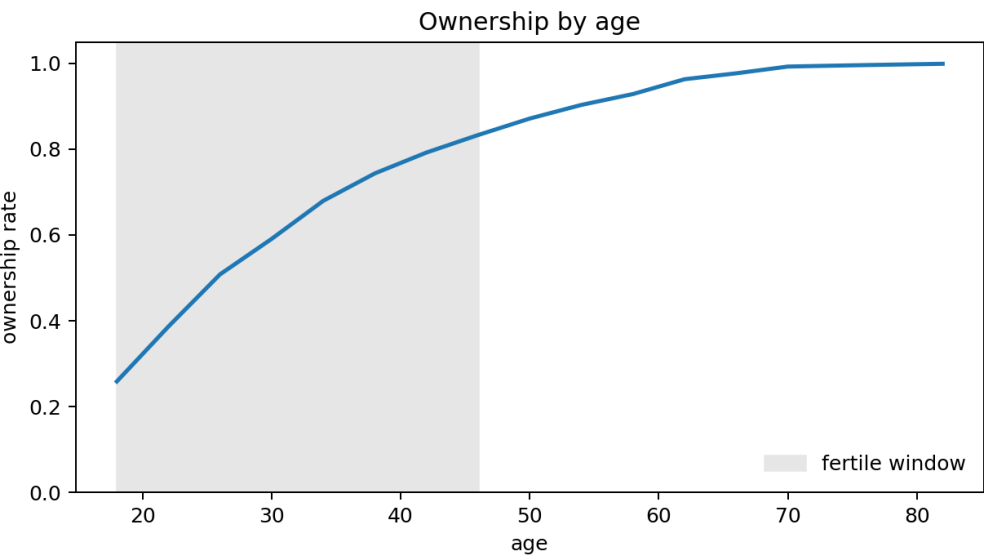
Floor, concave benefit



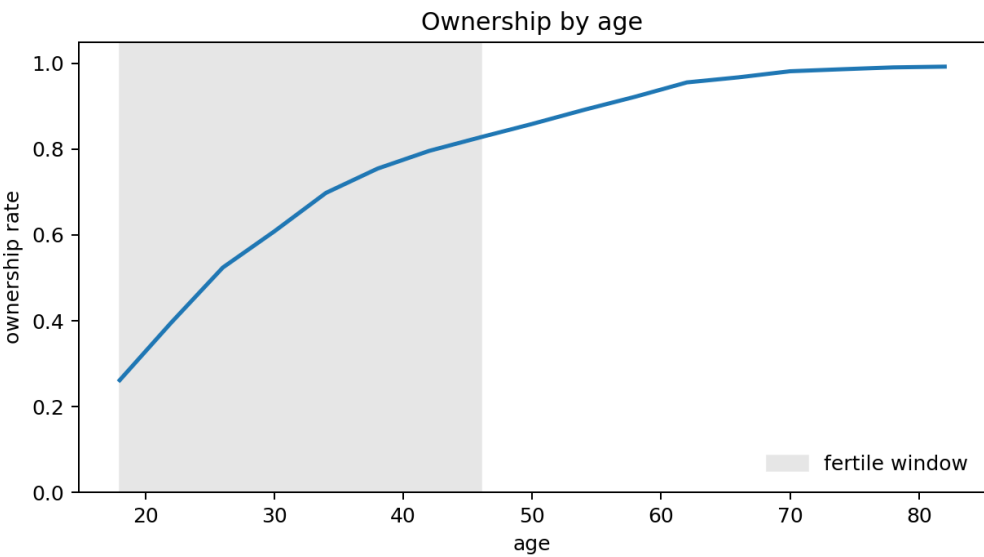
ownership by age

Standard diagnostic; same graph set for every completed case

No floor, no loading



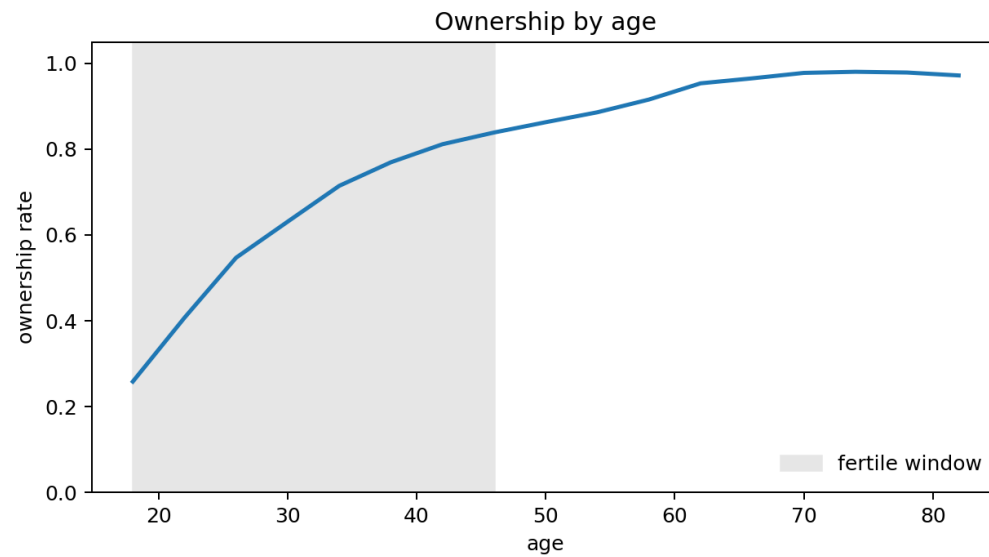
No floor, +10 points



ownership by age

Standard diagnostic; same graph set for every completed case

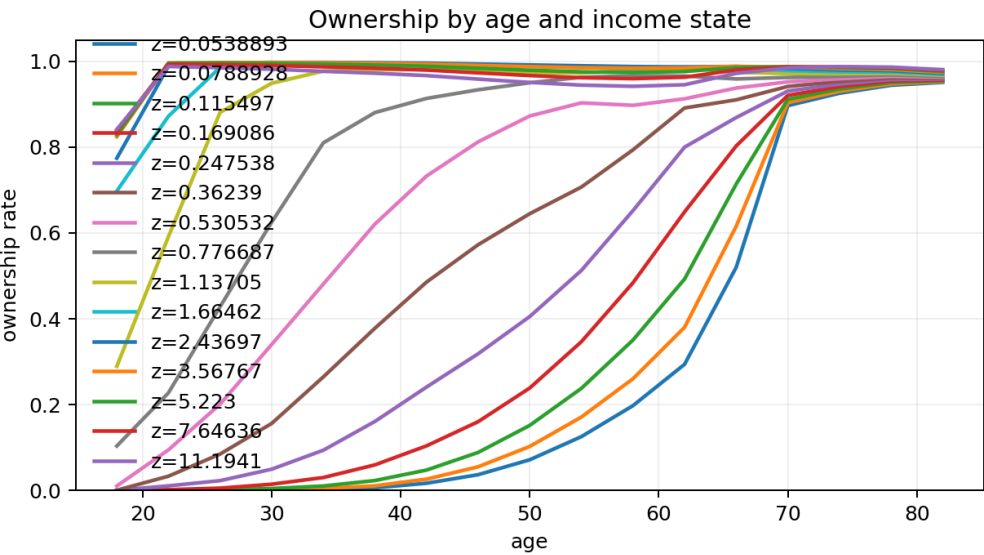
No floor, +20 points



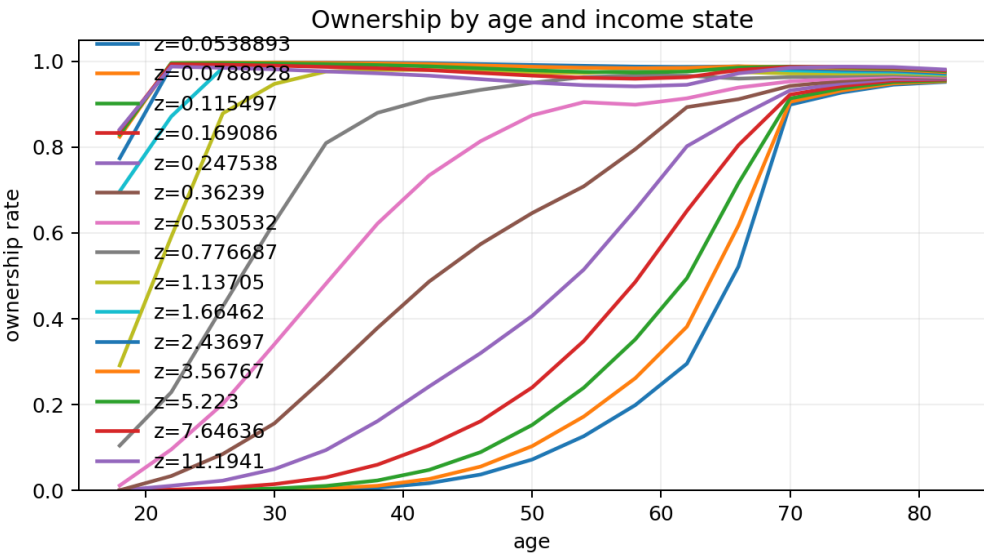
ownership by age income state

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



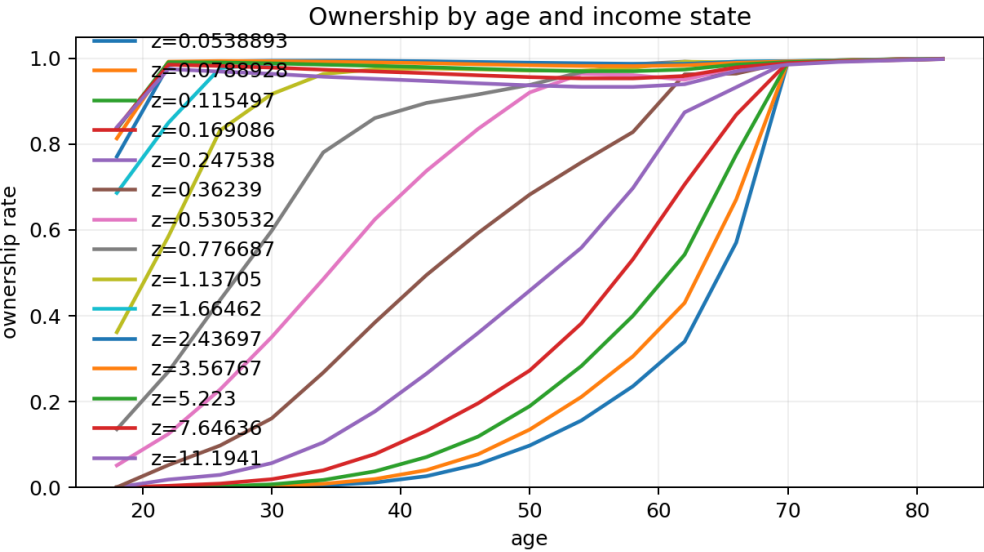
Floor, concave benefit



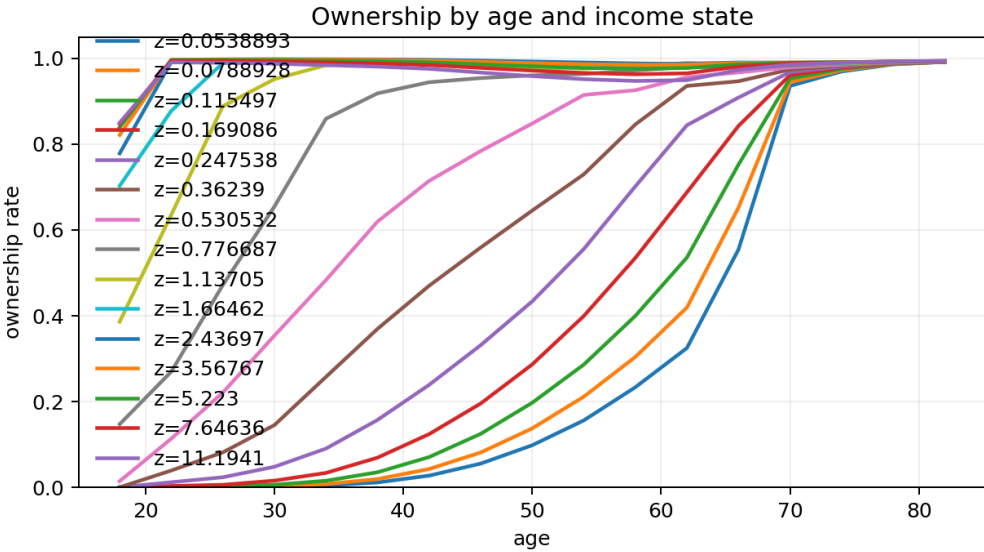
ownership by age income state

Standard diagnostic; same graph set for every completed case

No floor, no loading



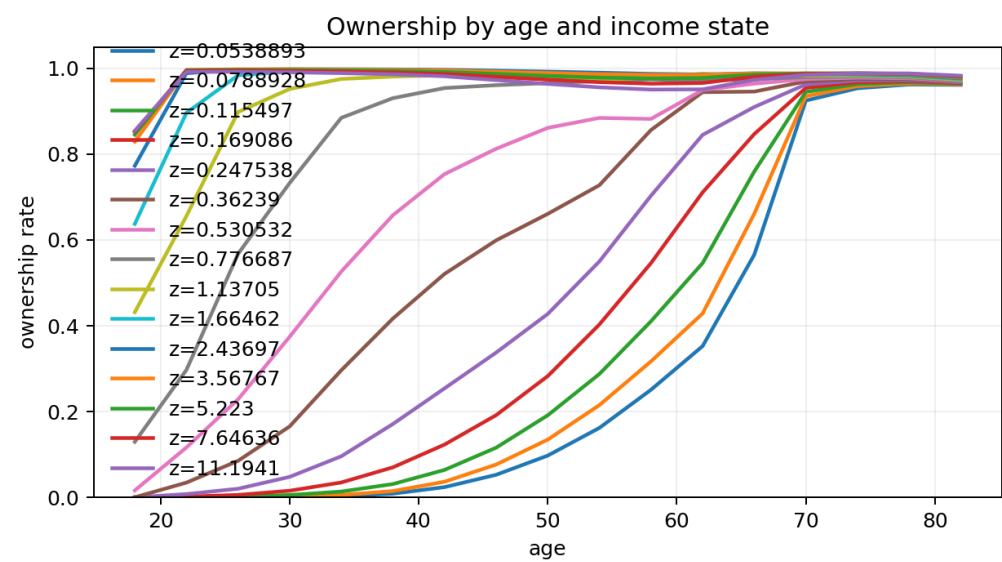
No floor, +10 points



ownership by age income state

Standard diagnostic; same graph set for every completed case

No floor, +20 points

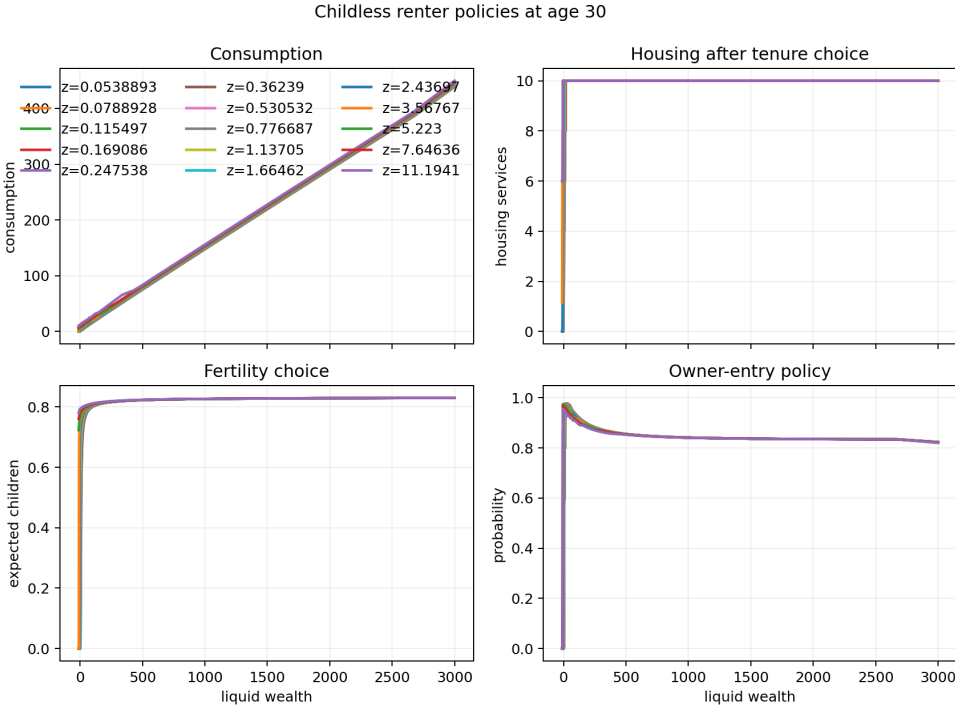
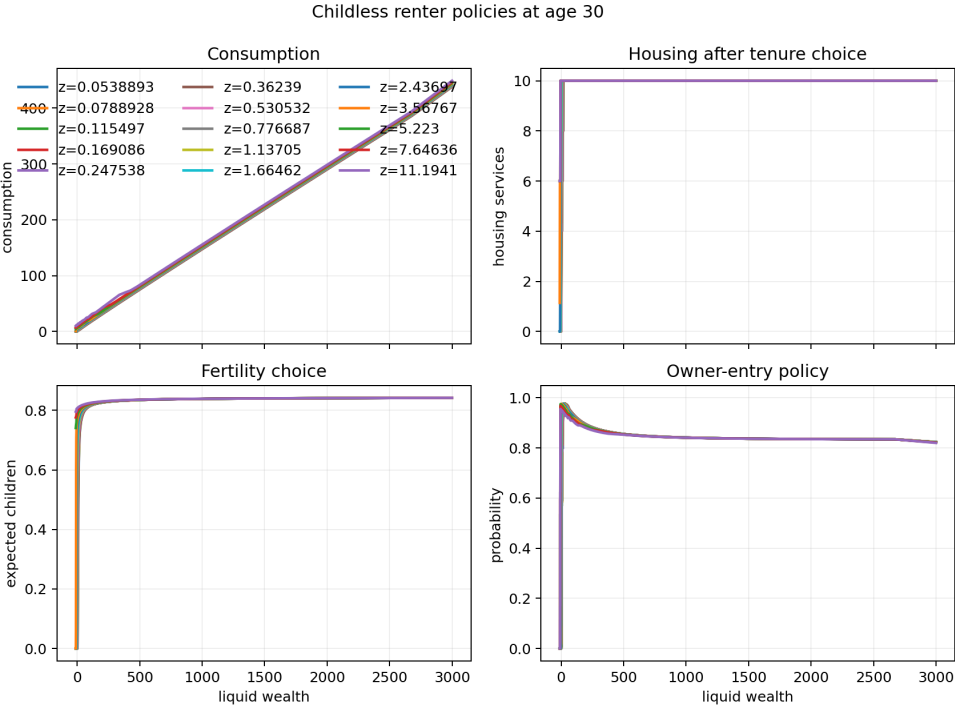


policy childless renter age30

Standard diagnostic; same graph set for every completed case

Floor, linear benefit

Floor, concave benefit

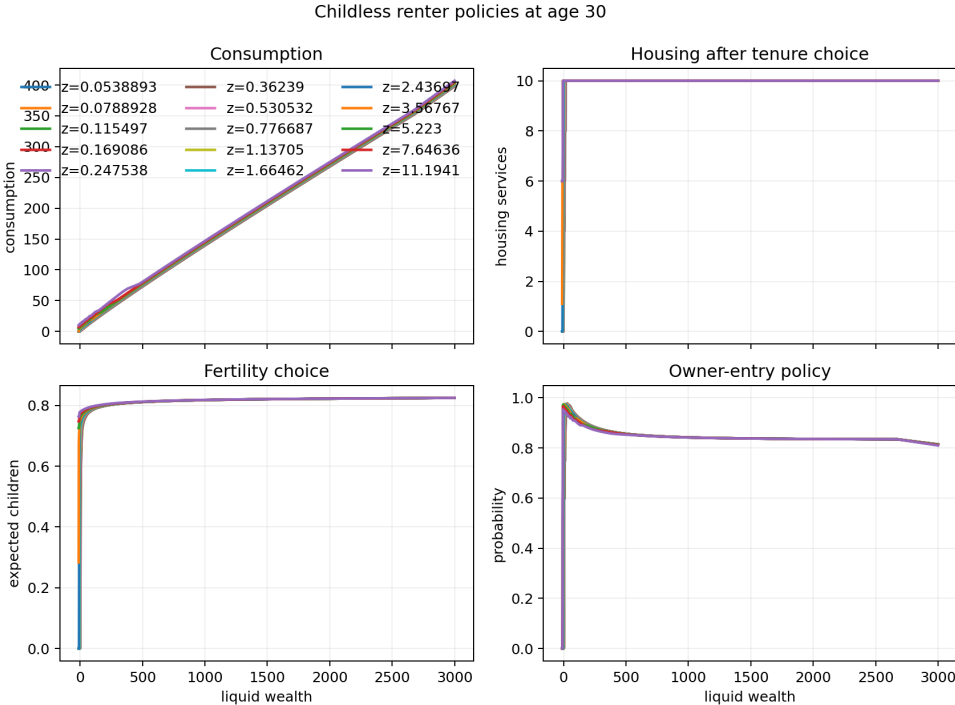
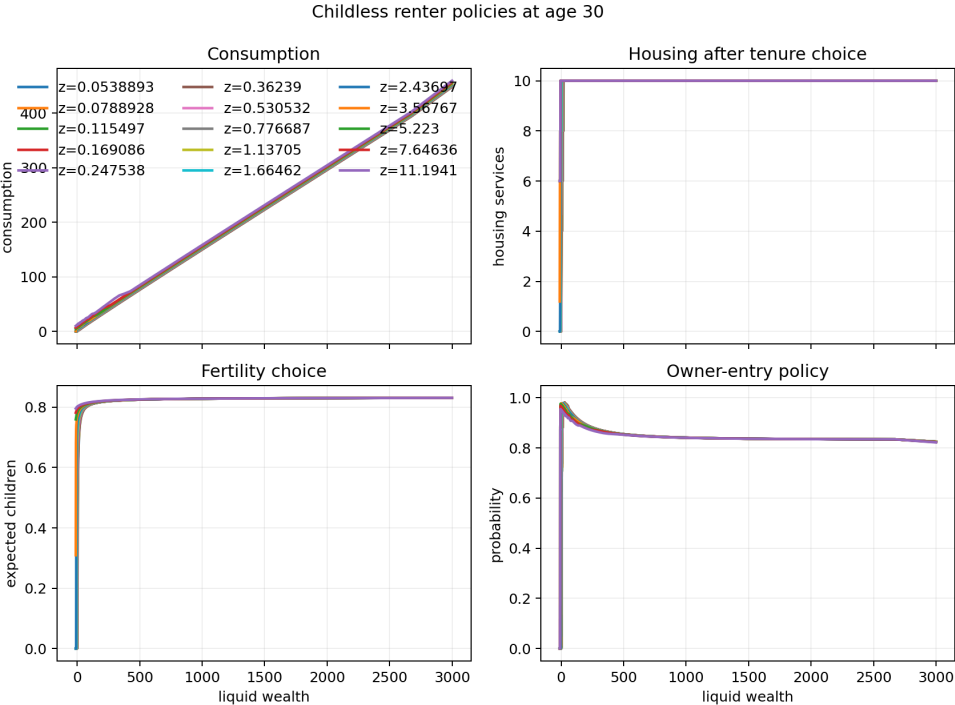


policy childless renter age30

Standard diagnostic; same graph set for every completed case

No floor, no loading

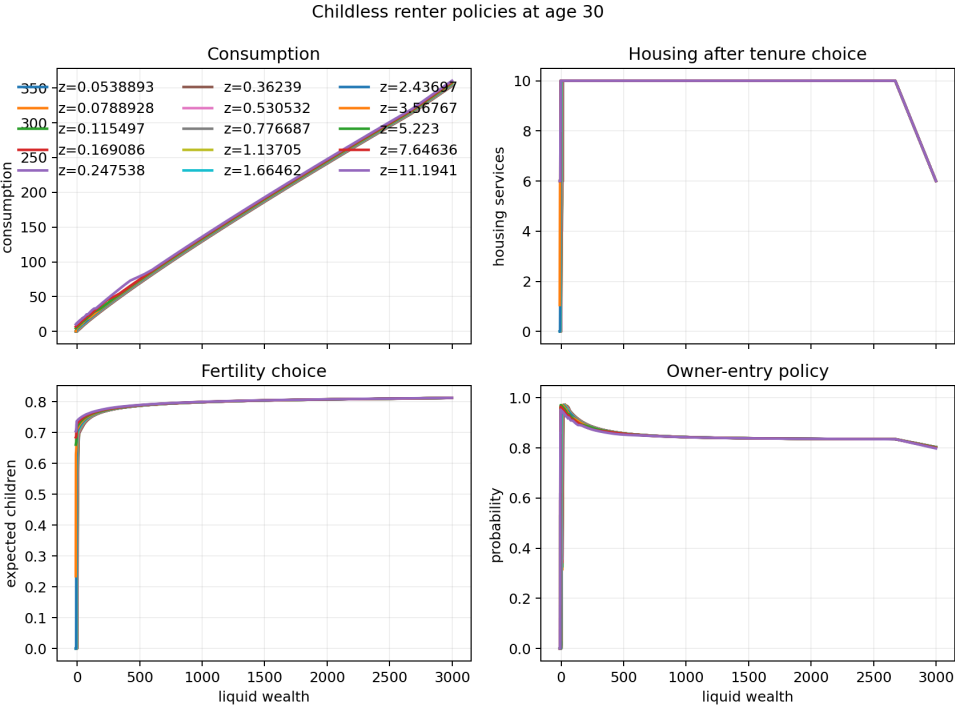
No floor, +10 points



policy childless renter age30

Standard diagnostic; same graph set for every completed case

No floor, +20 points

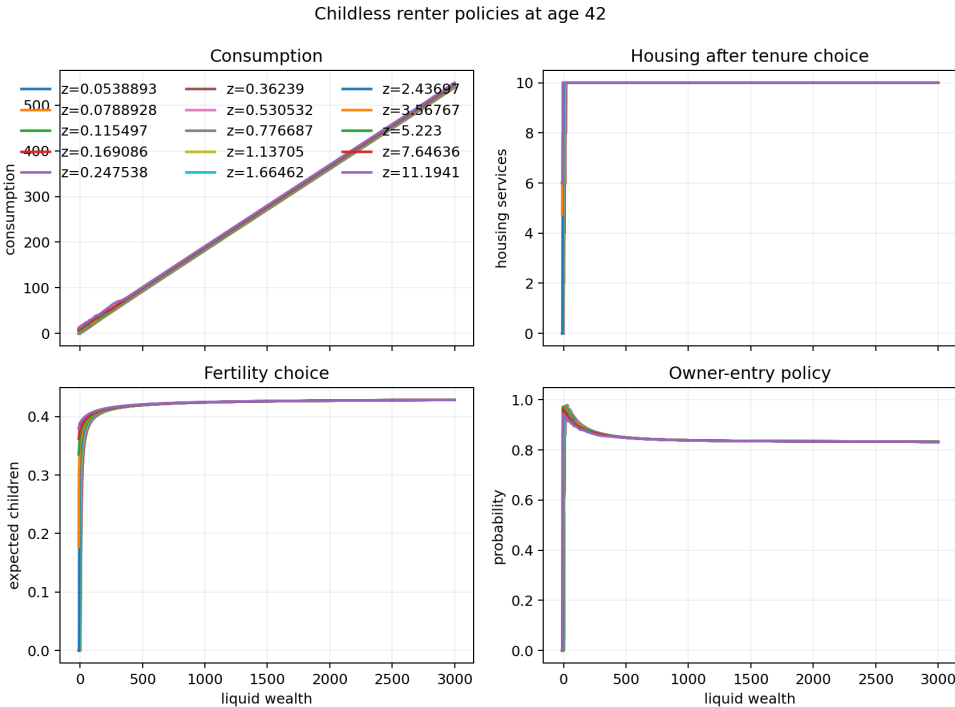
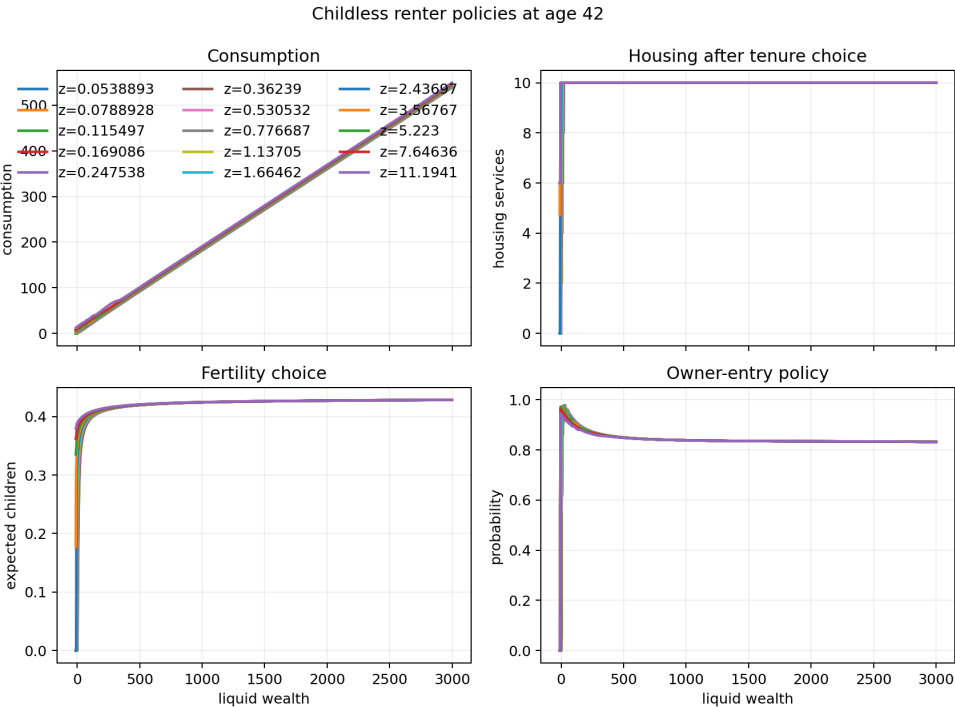


policy childless renter age42

Standard diagnostic; same graph set for every completed case

Floor, linear benefit

Floor, concave benefit

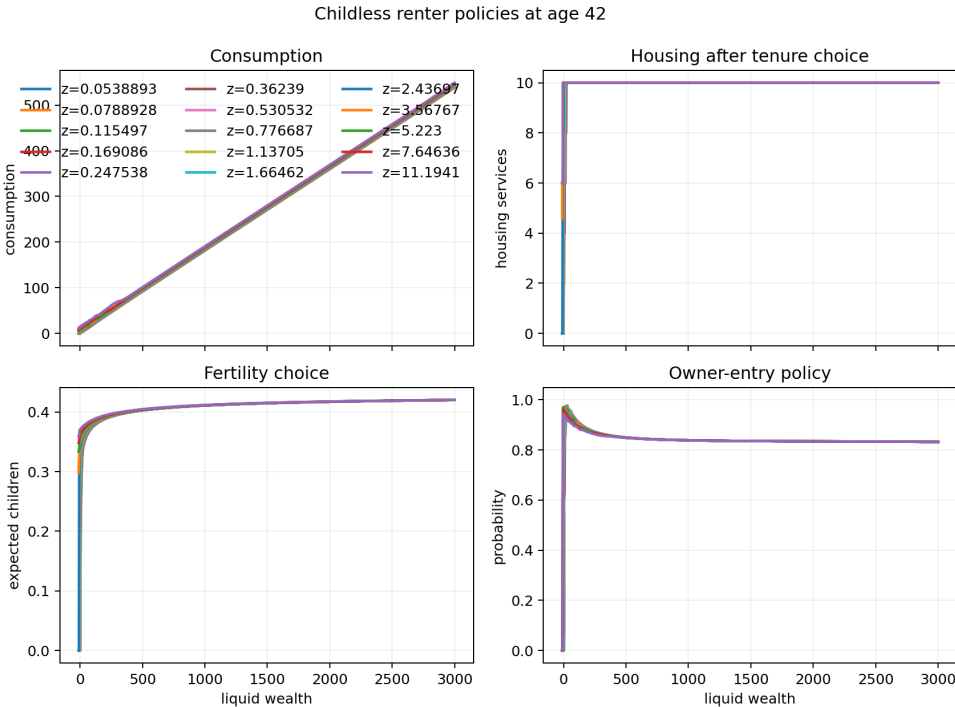
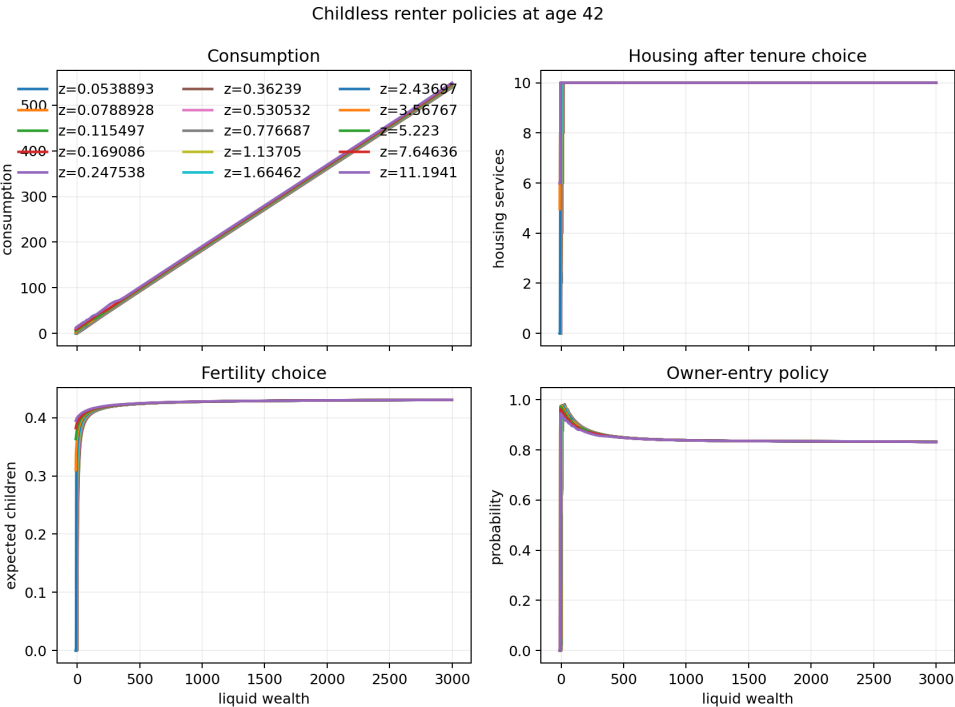


policy childless renter age42

Standard diagnostic; same graph set for every completed case

No floor, no loading

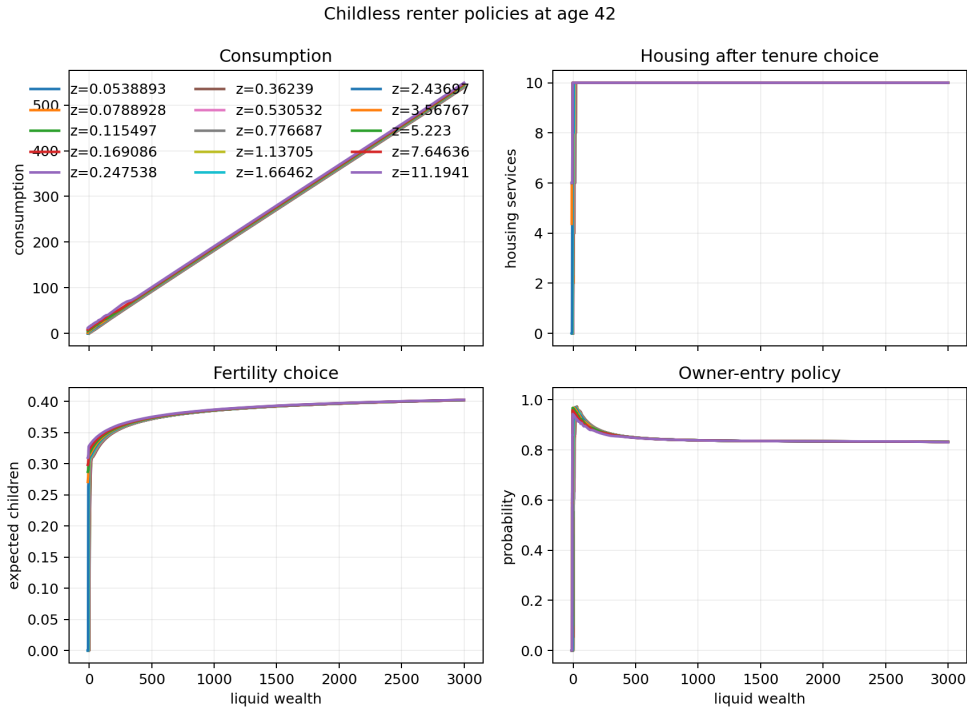
No floor, +10 points



policy childless renter age42

Standard diagnostic; same graph set for every completed case

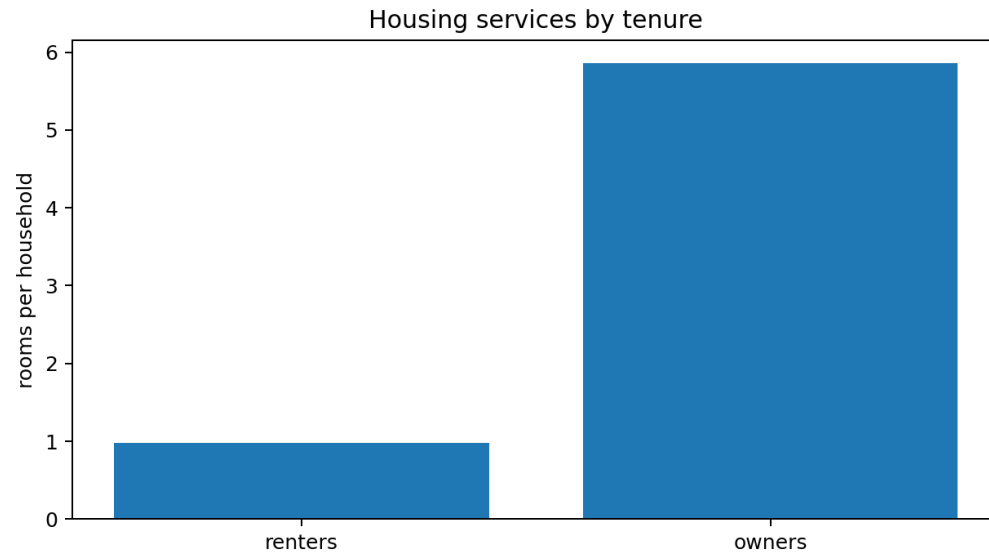
No floor, +20 points



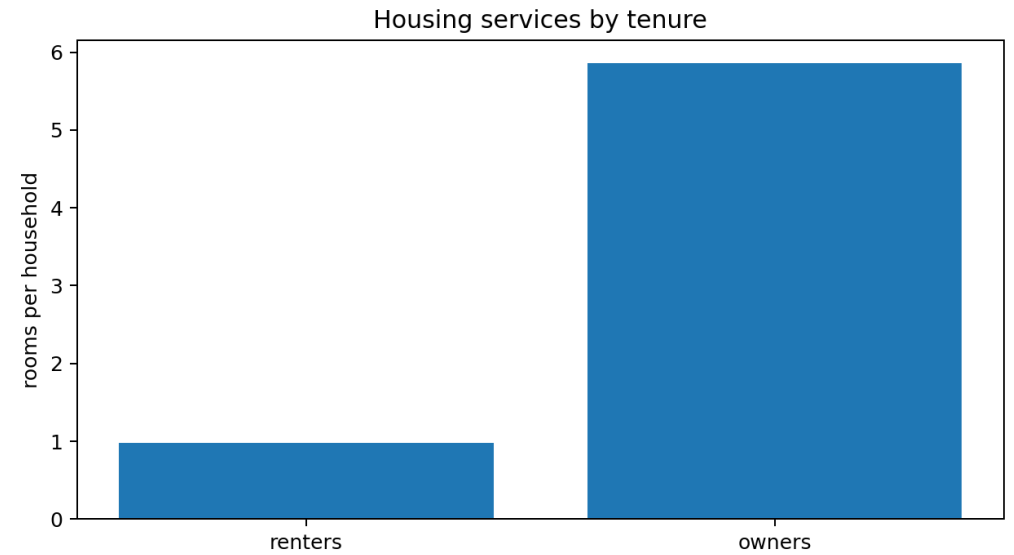
tenure services

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



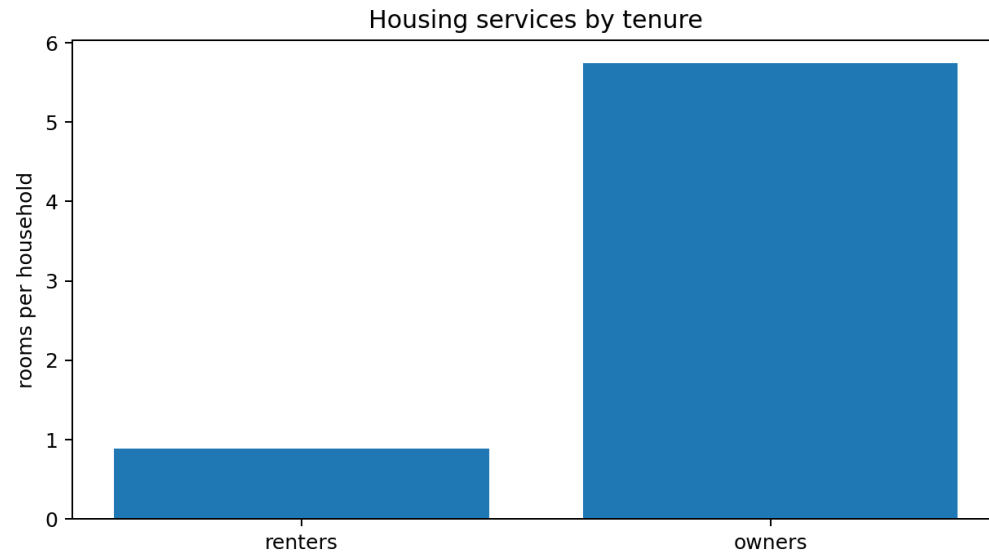
Floor, concave benefit



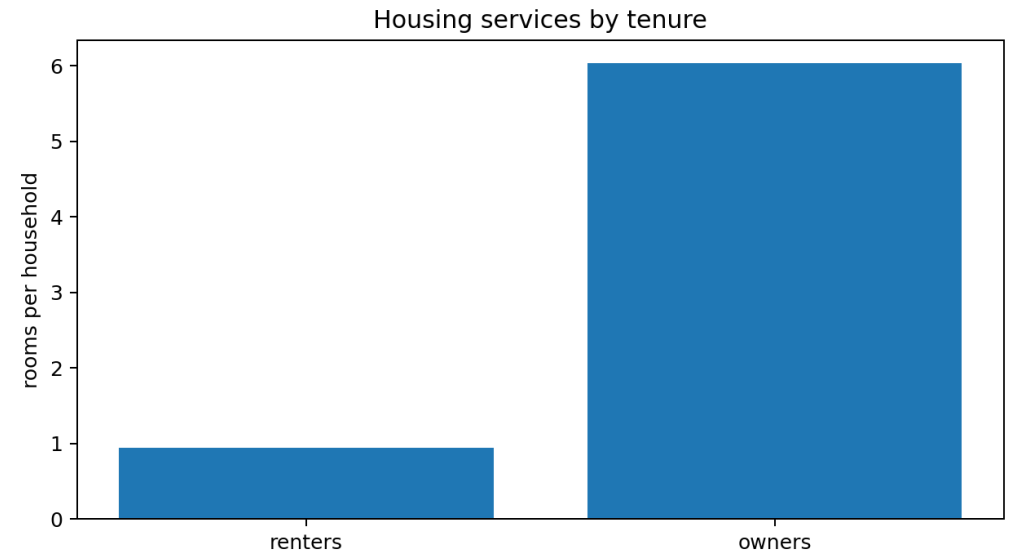
tenure services

Standard diagnostic; same graph set for every completed case

No floor, no loading



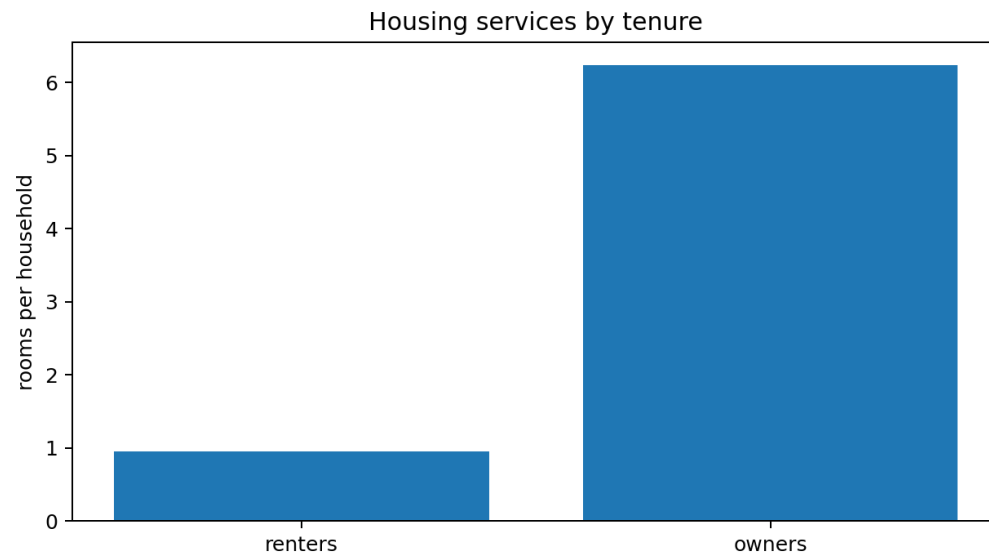
No floor, +10 points



tenure services

Standard diagnostic; same graph set for every completed case

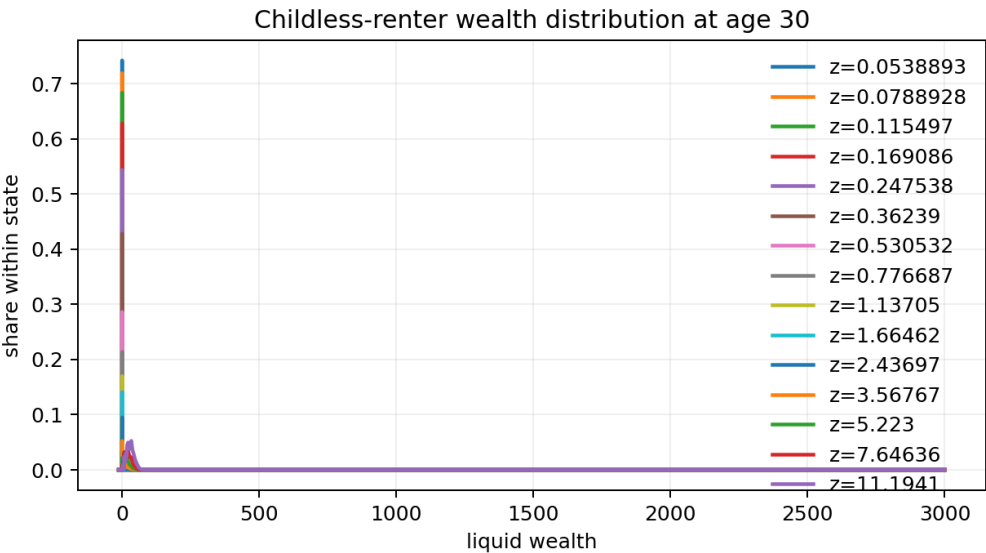
No floor, +20 points



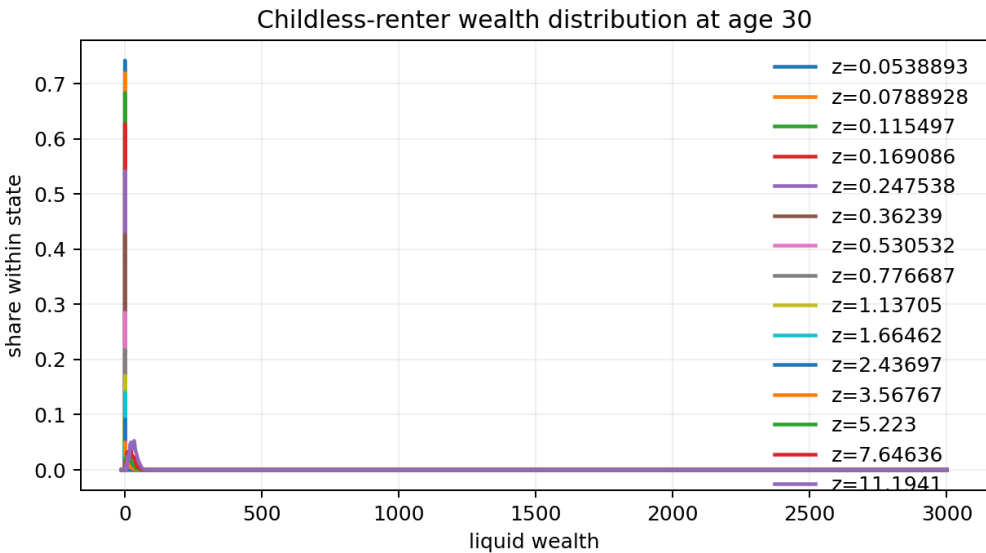
wealth dist childless renter age30

Standard diagnostic; same graph set for every completed case

Floor, linear benefit



Floor, concave benefit

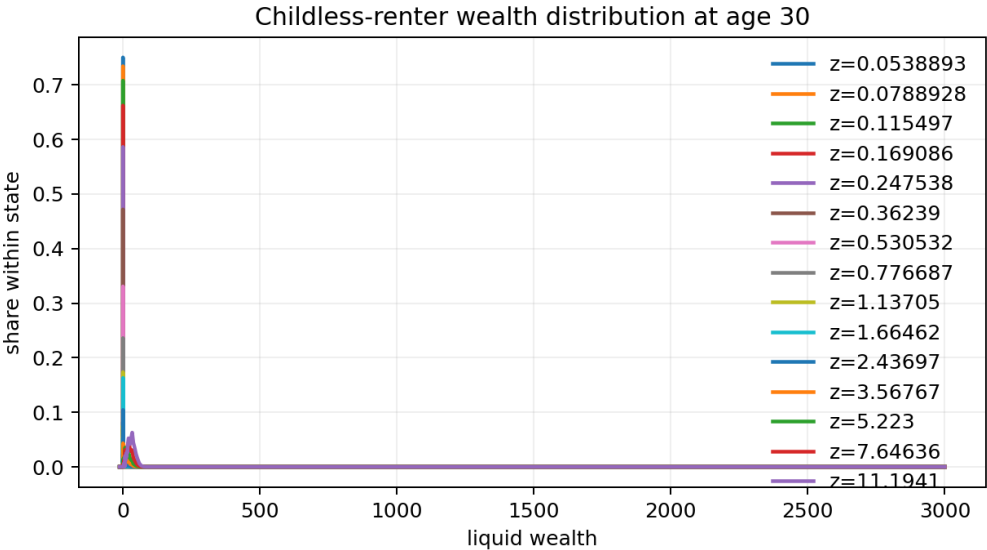
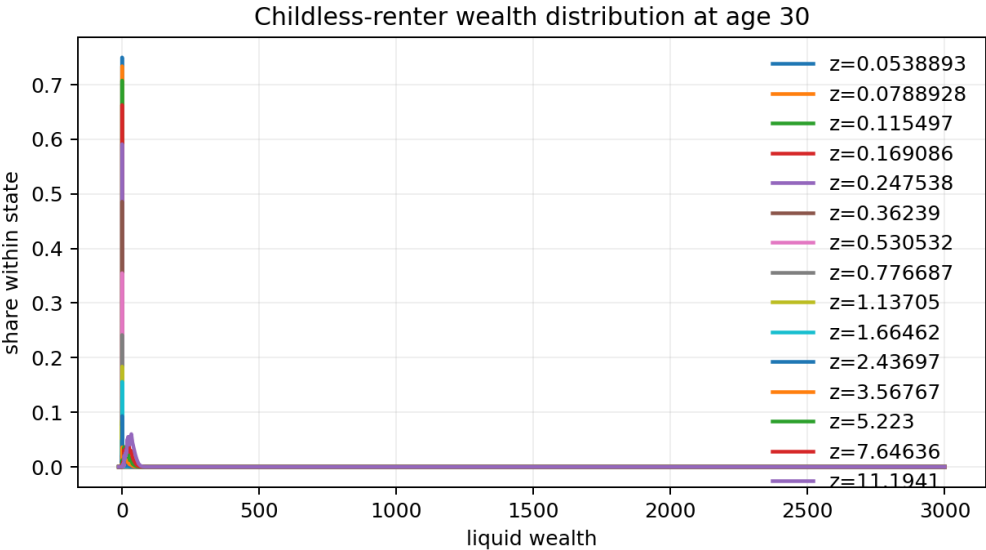


wealth dist childless renter age30

Standard diagnostic; same graph set for every completed case

No floor, no loading

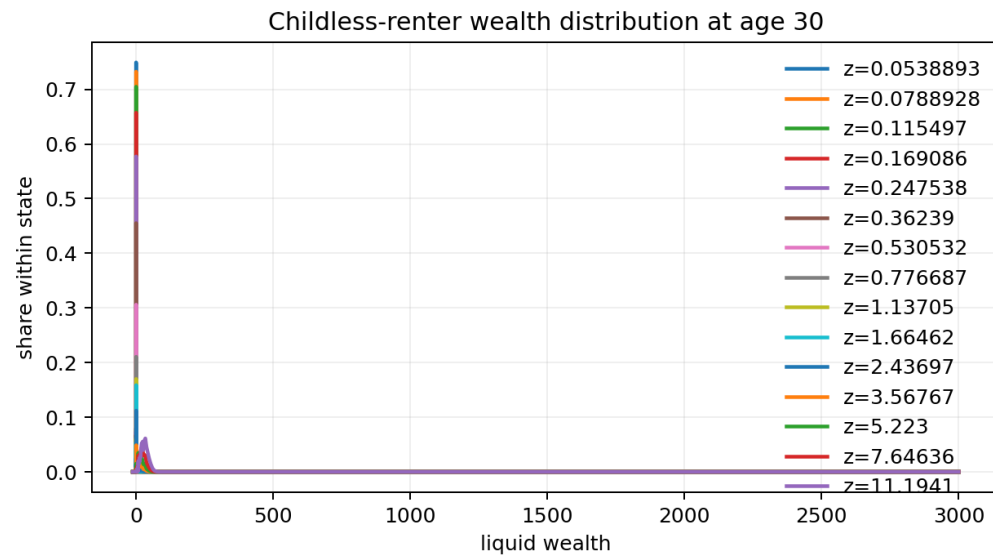
No floor, +10 points



wealth dist childless renter age30

Standard diagnostic; same graph set for every completed case

No floor, +20 points

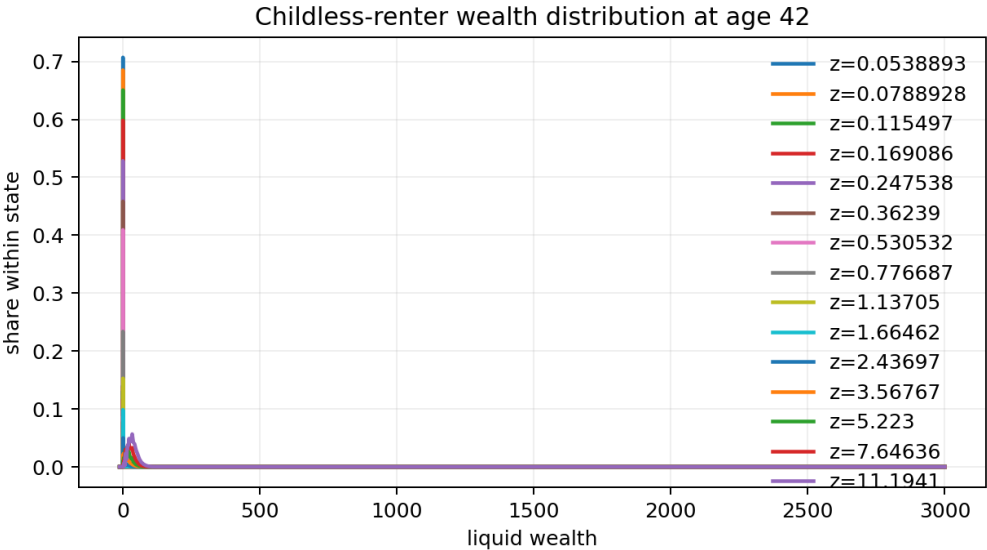
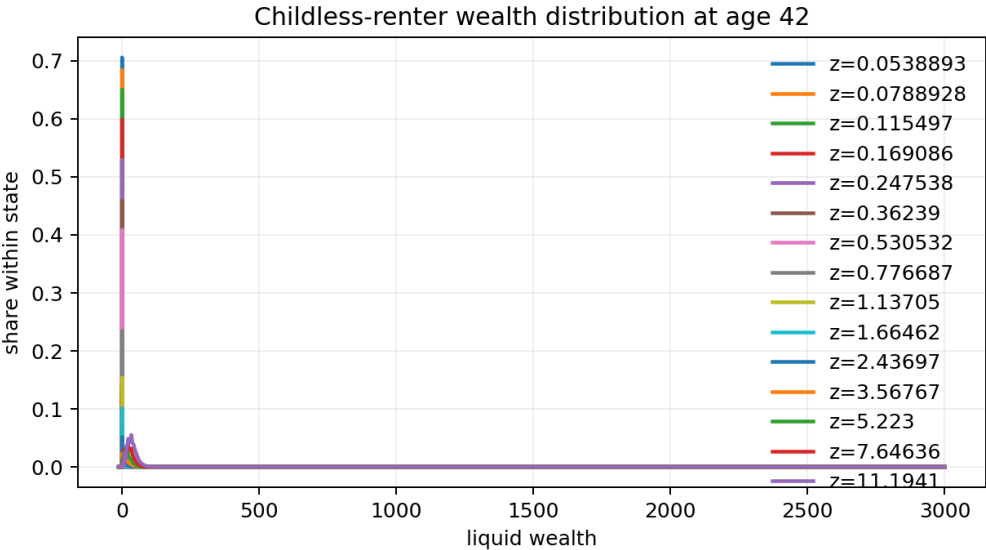


wealth dist childless renter age42

Standard diagnostic; same graph set for every completed case

Floor, linear benefit

Floor, concave benefit

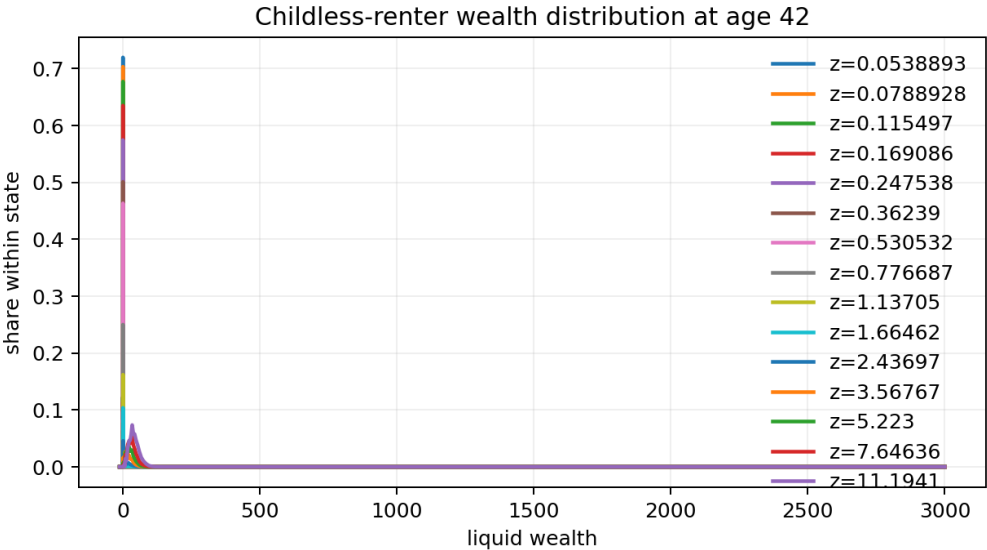
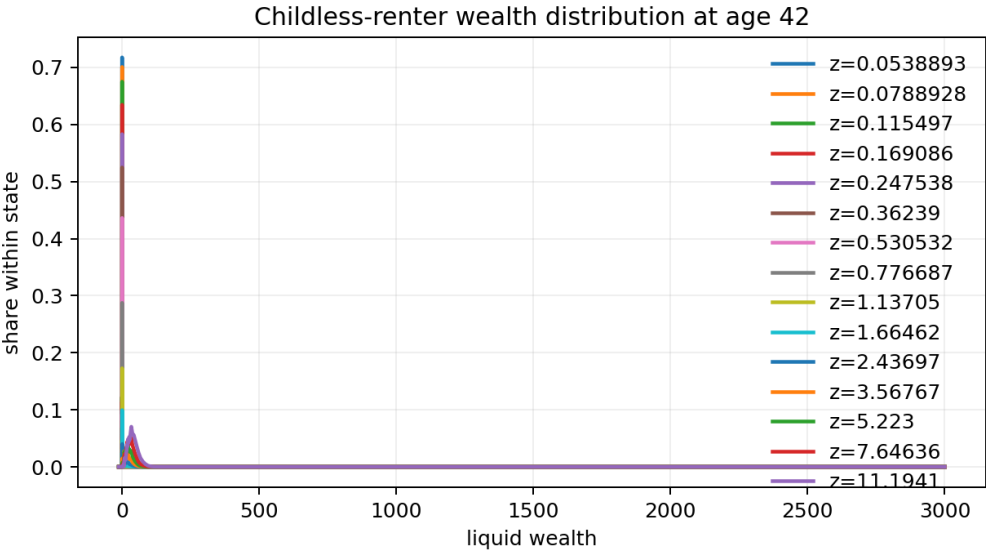


wealth dist childless renter age42

Standard diagnostic; same graph set for every completed case

No floor, no loading

No floor, +10 points



wealth dist childless renter age42

Standard diagnostic; same graph set for every completed case

No floor, +20 points

